RESEARCH REPORT

Allocating AI Translation Compute for Marginal Educational Access

A global language, interlanguage, accessibility, and open-curriculum portfolio model

A global portfolio model for language, shared-core, accessibility, and open-curriculum decisions

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Abstract

Artificial intelligence sharply reduces some costs of adapting open educational resources, but "translate into the largest language" is not an adequate allocation rule. A useful portfolio must ask how many people gain *new, comfortable, usable* access; which content is absent; whether another edition or academic lingua franca already serves the same readers; whether the output works in the relevant script, device, bandwidth, and disability context; and how much source preparation, terminology work, mathematical checking, correction, and maintenance the edition requires. This paper constructs a reproducible global allocation model around that marginal-access question. It joins dated language-population observations to exact language/variety/script/territory interventions, local Open Logic and OpenStax workloads, advanced-material scarcity proxies, accessibility requirements, and evidence-bounded interlanguage architectures. The common language-priority comparator is a 210-unit, 120,083-source-token formal-reasoning curriculum; larger OpenStax portfolios are costed separately so that useful breadth is not mistaken for inefficiency. Gross demographic ceilings, evidence-backed marginal floors, and wide neutral sensitivities are reported separately. Constructed bridges receive no blanket family reach; shared semantic cores receive compute savings only through explicitly named localized outputs. The source register contains 474 population observations from 80 authorities; 142 exact natural-language/profile interventions were scored and 133 passed the cardinal eligibility gate. Every direct-evidence marginal lower bound is zero, so the headline portfolio alternates explicitly labelled base, optimistic, and scarcity sensitivities rather than fabricating a strict conservative order. Its Top 10 is Indian Bengali, Bangladesh Bangla, Telugu, Marathi, Vietnamese, Javanese, Indian Tamil, Western Punjabi, Gujarati, and Kannada. Producing the 100 selected formal-reasoning editions is modeled at 91.274 million, 407.305 million, or 1.866 billion gross tokens under the low, base, and high workflows. Four complete shared-core architectures show modeled base-scenario savings of 25%–37.5%, but remain noncardinal because their frozen population-link controls do not authorize additive package unions. Eleven accessibility interventions are preserved in a separate safeguard backlog because their linguistic and disability/connectivity population ceilings overlap. The result is a grant-oriented commissioning order with exact production identities, machine-readable provenance, and uncertainty that remains visible rather than being converted into invented precision.

Keywords: educational access; language of instruction; open educational resources; machine translation; low-resource languages; interlanguages; OpenStax; Open Logic Project; accessibility; cost-effectiveness

1. Introduction

The relevant optimization problem is not how many translated files an AI system can produce. It is how much *additional usable educational access* a finite amount of compute can create. Those quantities diverge whenever a nominal language edition is not readable in the relevant variety or script, duplicates comfortable access through another language, contains content above the learner's current level, fails to reach low-bandwidth or disabled readers, or remains detached from instruction and actual use.

Two widely cited global language-of-education statistics illustrate why the denominator must be defined before it is optimized. Walter and Benson's (2012) Table 14.2 (p. 283) estimated that 3,741,110,588 people belonged to languages used in formal education and 2,300,263,716 to languages not used, yielding 61.9% and 38.1% of their covered global population. The same table reported 97 languages with more than ten million speakers: 52 used in education and 45 not used. Those 45 accounted for 1,120,220,125 people, almost half of the population in the table's “not used” side. The chapter prints size bands, not a list of the 97 languages. It therefore supports a concentration argument—large neglected languages matter—but cannot identify a modern target list.

The World Bank's later estimate is different. It reports that approximately 37% of students in low- and middle-income countries, about 370 million of roughly one billion students, are taught in a language other than their first language or the language they best speak and understand (World Bank, 2021). This is a student and instructional-language estimate, not Walter and Benson's global language-population calculation. Neither percentage can be multiplied by every named language population to manufacture the number a translation would reach.

The underlying distribution nevertheless makes allocation consequential. In Walter and Benson's (2012) table, the 45 noneducation languages above ten million and the 662 in the 250,000–9,999,999 band together represented 2,088,576,471 people—90.8% of the population assigned to languages not used in education. A compute program can thus combine two aims rather than choose between them: large exact-language editions can deliver enormous aggregate reach, while smaller endangered, signed, Indigenous, and minoritized-language editions can expand public function, regional depth, and prestige domains that a population-only ranking erases.

AI changes the feasible frontier but not the definition of access. Multilingual models, terminology constraints, deterministic source pipelines, independent model critique, and formula/structure checks make many editions cheaper and more auditable. Yet low-resource model performance, corpus quality, dialect coverage, and specialist terminology remain uneven (Blasi et al., 2022; Dinu et al., 2019; Kantharuban et al., 2023; Kreutzer et al., 2022; NLLB Team, 2024; Robinson et al., 2023). A global portfolio therefore needs exact target identities, an explicit overlap model, and uncertainty that remains visible rather than being compressed into an opaque score.

2. Research questions

The study asks:

1. Which exact language, variety, script, territory, and accessibility interventions have the largest dated source-population ceilings among communities not already represented by the local Open Logic/OpenStax portfolio?
2. How do those ceilings change after directly evidenced or bounded deductions for literacy/modality access, academic-lingua-franca access, and existing editions?
3. Which Open Logic and OpenStax curriculum units provide the most defensible first educational package for each population, rather than a vague promise of "educational materials"?
4. How many low/base/high inference tokens and API-equivalent dollars would those packages require under a reproducible multi-pass translation and audit workflow?
5. When can an interlanguage, pluricentric semantic core, dual-script package, or natural-intercomprehension workflow improve reach per compute relative to separate editions?
6. Which exact natural-language or shared-core interventions form a deduplicated Top 10 and Top 100 under reach-first, scarcity, and overlap sensitivities, and which accessibility mechanisms must be retained as separate horizontal safeguards because their residual beneficiary counts cannot yet be ranked cardinally?

3. Conceptual framework

3.1 Access is not learning, but it is a necessary input

Language adaptation can remove a barrier without guaranteeing learning. A translated edition must still be instructionally appropriate, delivered, used, and integrated with practice. This paper therefore ranks **expected marginal educational access** rather than test-score effects. Learning-adjusted benefits require a specified deployment and appear only as a later scenario.

This choice also follows comparative cost-effectiveness practice: a common numerator is useful only when its outcome definition and cost boundary remain explicit (Angrist et al., 2025; Dhaliwal et al., 2013). Here the common numerator is access created, not a synthetic learning-effect estimate.

For population cell c, intervention i, and curriculum unit k:

$$\mathrm{EMA}_{cik} = \mathrm{N}_c\ \mathrm{D}_{ck}\ \mathrm{C}_{ci}\ \mathrm{P}_{ci}\ \mathrm{U}_{ci}\ \mathrm{R}_{ck},$$

where N is a dated source population; D is the missing-content share; C is comprehension of the proposed linguistic surface; P is practical written, signed, device, bandwidth, and disability access; U is residual non-overlap; and R is relevance of the fixed curriculum to the specified learner stratum. Scarcity, vitality/equity, prestige-domain value, production feasibility, and dialect-flattening risk remain separate views rather than invisible multipliers of people.

This extends the recovered local marginal-intelligibility formalism (*Crosswalk to earlier local marginal-intelligibility research*, n.d.; *Marginal-intelligibility opportunity heuristic for OpenLogic*, n.d.). Its core opportunity dimensions are newly comfortable readers (R), scarcity (S), practical accessibility (A), and non-overlap (N), with vitality/prestige/feasibility/risk kept separate. Numeric intervals refine those bands without redefining them.

3.2 Three quantities that must not be conflated

Every language table distinguishes:

- **source observation:** exactly what the source counted, including year, unit, and measure type;
- **new-edition access ceiling:** the compatible written or modality population for an exact target; and
- **marginal comfortable-access range:** the residual after access through existing editions and academic lingua francas is deducted.

A mother-tongue census count, a household-language count, self-reported speaking ability, an ethnic-group population, a refugee caseload, and school enrolment are not interchangeable. Aggregate Quechua, Nahuatl, Maya, Cree, Ojibwe, Inuktut, Fula, Kurdish, Amazigh, or Romani labels are not assigned to a single edition surface.

3.3 Interlanguage mechanisms

"Interlanguage" covers mechanisms with different numerators and denominators:

1. A **constructed bridge surface** seeks receptive comprehension across named communities. It earns reach only from evidence for that surface, script, cohort, and task.
2. A **pluricentric shared semantic core** generates explicitly named localized outputs. Reach belongs to each native-language output; the shared source reduces compute.
3. **Natural intercomprehension reuse** shares terminology, source structure, and QA across related languages while preserving separate editions.
4. A **dual-script/register package** produces synchronized outputs for one language continuum without adding script populations twice.
5. An **accessibility derivative** removes a format, device, register, or disability barrier within a language population and never substitutes for linguistic access.

This taxonomy prevents a family name from turning into hundreds of millions of unsupported readers. The current Interslavic evidence includes an approximately .84 pooled result on a seven-gap short written cloze task, but no low/high interval and no demonstration of sustained mathematical reading. The pooled point is retained as an upper-bound research sensitivity; it is not inherited by Russian, Ukrainian, Polish, Czech, Slovak, South Slavic, or minority-language leaves (Merunka et al., 2019).

4. Evidence from education, OER, AI, and accessibility

4.1 Familiar-language instruction can help, with heterogeneous effects

Nakamura et al.'s (2023) systematic review included 45 studies and found positive pooled effects for several literacy outcomes: letter knowledge, sentence reading, reading comprehension, and national-language reading comprehension. Word reading was near zero, and writing evidence rested on one study. The result supports language as an enabling condition, not one transportable effect size.

Setting-specific evidence shows both upside and implementation risk. A Kom-medium program in 24 Cameroonian schools reported large early gains and persistent later gains after transition to English (Laitin et al., 2019). South African administrative panel evidence associated early transition to English with lower later English achievement than three years of home-language instruction (Taylor & von Fintel, 2016). Conversely, a Kenyan trial adding Lubukusu or Kikamba literacy to an English/Kiswahili package did not improve the wider-language outcomes and produced lower mathematics results in that implementation; limited classroom use, teacher language mismatch, and resistance mattered (Piper, Zuilkowski, Kwayumba, et al., 2018).

One official language-specific administrative example shows why the global 37%-40% figures cannot simply be assigned to every language. South Africa's 2016 school tabulation reported 2,402,540 Foundation Phase learners whose school offered a language of learning and teaching matching home language and 540,484 whose school did not; in Intermediate Phase the corresponding counts were 585,808 and 1,844,086 (Department of Basic Education, 2023). These are phase-specific principal-reported EMIS cells with verification warnings, not a global causal estimate or a count that can be multiplied by the 2022 census. They do, however, identify a concrete cohort and show how medium-of-instruction mismatch can rise sharply after the early grades.

The allocation consequence is precise: familiar-language evidence justifies a linguistic-usability term and exact cohort research. It does not justify multiplying a static PDF by a generic standardized-effect coefficient.

4.2 Complete instructional packages outperform file production

The clearest materials evidence favors structured pedagogy. In a Kenyan randomized evaluation, professional development, coaching, student books, and structured teacher guides operated as a package and produced large gains; teacher guides added little cost relative to the other components in that setting (Piper, Zuilkowski, Dubeck, et al., 2018). McEwan's (2015) meta-analysis found positive average effects across several intervention classes but also showed that composite treatments and sparse cost data complicate category rankings.

Input-only failures matter. Textbooks in 100 Kenyan schools had no average effect and benefited mainly higher-performing pupils; the books were in English and pitched to a demanding curriculum (Glewwe et al., 2009). In Sierra Leone, many supplied books remained stored and no learning effect was detected (Sabarwal et al., 2014). The Global Education Evidence Advisory Panel (2023) accordingly treats structured pedagogy and teaching at the right level more favorably than unintegrated inputs.

Each recommended edition therefore names a bounded curriculum and derivative set: editable source, preserved examples/exercises, solution material where licensed, teacher sequence or misconception guidance where available, diagnostic/self-check material, accessible HTML/MathML, visually checked PDF, and offline distribution.

4.3 Open licensing creates option value, not an automatic learning effect

Open textbook meta-analysis finds little average achievement difference between open and commercial textbooks, with some evidence of lower withdrawal and strong access and cost advantages (Clinton & Khan, 2019). Reviews and observational studies are generally compatible with "no worse learning, lower access barriers," but most are not randomized and are concentrated in North American higher education (Colvard et al., 2018; Hilton, 2016).

Grimaldi et al. (2019) formalize the access hypothesis: when only students previously unable to obtain a commercial text benefit, whole-class averages dilute the effect. That mechanism motivates the present numerator—newly usable access rather than all speakers, downloads, or enrolments. UNESCO's (2019) OER Recommendation supplies an international policy framework encouraging open licensing, adaptation, translation, redistribution, and accessible formats; permission to adapt a particular work depends on its applicable license. Openness enters this model as reuse and future-option value, not a test-score multiplier.

4.4 Model coverage is not source-faithful specialist translation

NLLB demonstrates the engineering value of multilingual transfer, but aggregate benchmark gains coexist with large resource-level disparities (NLLB Team, 2024; Robinson et al., 2023). Corpus audits show why nominal data volume can mislead: Kreutzer et al. (2022) found that many web-crawled language corpora contained less than half usable text and some contained no in-language material in the audited sample. Dialect performance gaps are substantial and inconsistent (Kantharuban et al., 2023); relatively small dialect-specific parallel corpora have produced large improvements over standard-language-oriented systems in earlier MT settings (Zbib et al., 2012).

Terminology constraints raise exact term-use rates but do not guarantee grammar, semantics, or mathematical fidelity (Bergmanis & Pinnis, 2021; Dinu et al., 2019). The compute model therefore budgets initial translation, terminology work, mathematical/source comparison, structural QA, build/render inspection, correction cycles, accessibility derivation, and retry recovery. Benchmark inclusion contributes to feasibility; it never becomes linguistic authority.

4.5 Accessibility and low bandwidth are separate marginal-access axes

WCAG 2.2, EPUB Accessibility 1.1, and MathML Core provide machine-testable and judgment-dependent requirements for web, publication, and mathematical structure (W3C Math Working Group, 2025; World Wide Web Consortium, 2024a, 2024b). Conformance is not the same as successful use across every assistive-technology stack, but native semantic source avoids repeatedly remediating inaccessible PDFs.

Low-tech evidence again emphasizes intervention design. Phone tutoring plus SMS improved numeracy in Botswana, while SMS alone did not (Angrist et al., 2022). A Sierra Leone phone-tutoring intervention did not improve language or mathematics amid limited fidelity and take-up (Crawfurd et al., 2023). Offline HTML/EPUB, downloadable packages, audio, plain language, Braille, and signed-language video are therefore costed as specific derivatives, not credited merely because a file can be sent over a low-bandwidth channel.

5. Data

5.1 Population observations

The population master preserves source labels, years, measure definitions, units, territories, exact counts or source intervals, nesting, and alternative-measure groups. Official censuses and statistical reports are preferred. Where an official source lacks direct language measurement, an ethnic group is retained only as a proxy/context row and cannot enter the language ranking. Survey case frequencies are not national populations. Same-name Indian C-16 mother-tongue components are used in preference to broader scheduled-language categories; C-17 English/Hindi knowledge remains a category-level overlap sensitivity.

Where a direct national count was not registered, selected large-language ceilings use the pinned CLDR 48.2 population XML and transparent derivation chain. CLDR's heterogeneous estimates have unspecified component vintages and intervals, so these rows remain secondary ceilings rather than census observations (Unicode Consortium CLDR Technical Committee, n.d., 2025, 2026b).

The final frozen inventory and region/source counts are inserted by the build:

Table 1. Frozen analytical inventory and cardinal treatment.

Registered layer	Rows	Cardinal treatment
Dated population observations	474	Source facts; heterogeneous measures are never summed globally
Population-source authorities	80	Provenance and limitation register
Candidate intervention hypotheses	209	Discovery universe, including unrankable hypotheses
Scored exact natural-language/profile rows	142	105 prior rows plus 37 globally balanced expansion rows
Eligible exact natural-language interventions	133	Headline cardinal inventory
Richly served negative controls	4	Excluded from generic translation priority
Unresolved or D0 profile exclusions	5	Three unresolved profiles and two population/profile mismatches
Intervention-matrix rows	104	80 interlanguage plus 24 accessibility rows; not automatically cardinal
Interlanguage population links	113	All remain nonadditive under the frozen upstream gate
Package × token × reuse sensitivity rows	135	Engineering comparison only; zero packages enter the headline ranking
Accessibility safeguard interventions	11	Separate noncardinal backlog; zero numbered linguistic positions

Note. Row counts describe registered records, not unique people. The 104 intervention-matrix rows include both interlanguage and accessibility mechanisms; the 80-row interlanguage subset is summarized separately in Appendix H.

5.2 Existing local editions and work in progress

The bounded successor-state census verifies complete Open Logic baselines for zh-Hans-CN, hi-Deva-IN, tr-TR, Modern Standard Arabic, fa-Arab-IR, Spanish, and pt-BR; the Chinese lane also has all 94 Algebra and Trigonometry 2e modules and three of 55 Calculus Volume 1 modules at its pinned state. It distinguishes these from partial Indonesian and Interslavic work and provisional/nonadmitted bridge artifacts. A complete exact edition is baseline coverage, not a new intervention. A partial edition receives only the measured missing-content share. For Indonesian, the current pinned checkpoint covers 321 of 722 Open Logic units, but those units contain 181,791 of 367,220 source alpha tokens. The 401-unit remainder contains 185,429 tokens, 50.495% of source content; completion cost is therefore based on tokens rather than the larger 55.540% unfinished unit share. Appendix D identifies the exact checkpoint, measurement table, byte count, and SHA-256 used for these calculations.

5.3 Curriculum denominators

The principal comparator is the complete **FR-2 Formal Reasoning Core** at D3: 210 editable units and 120,083 measured source alpha tokens. Independent remeasurement verified every one of 722 Open Logic source units and the complete 367,220-token closure. The pinned source identity is OpenLogic commit `1e960beff9ed7835bf3e3f1335e21af3439cd107`, licensed CC BY 4.0 unless a component states otherwise (Open Logic Project, 2026).

OpenStax supplies concrete expansion portfolios rather than a vague content category. The locally available measured set comprises 257 modules and 1,017,337 source tokens: Algebra and Trigonometry 2e (94 modules; 394,473), Calculus Volume 1 (55; 195,458), Calculus Volume 2 (54; 180,155), and Calculus Volume 3 (54; 247,251). Four further books have exact module counts but only planning token intervals and are never misrepresented as measured. Candidate recommendations choose among a minimum viable quantitative gateway, statistics/business, science bridge, undergraduate STEM, and formal-reasoning portfolios, plus accessibility derivatives. Appendices A and F identify the exact collection commits, licenses, row-level curriculum measurements, and compute artifacts behind every denominator (OpenStax, n.d.-a, n.d.-b, n.d.-c, n.d.-d, n.d.-e, n.d.-f, n.d.-g, n.d.-h).

5.4 Scarcity, digital context, and model support

OpenAlex counts of mathematical works and open-access books are retained as a bibliographic scarcity proxy (OpenAlex, 2026). Wikimedia project counts provide broad digital-presence context (Numberof bot, 2026). Country internet, electricity, literacy, and enrolment indicators provide delivery sensitivities (World Bank, 2026). FLORES/SONAR and NLLB resource levels provide model-support context (Duquenne et al., 2023; NLLB Team, 2024). None of these is relabelled as an inventory of usable advanced teaching material, a language-specific accessibility rate, or proof of translation quality.

6. Methods

6.1 Exact intervention edge construction

Machine joins use exact ISO/code crosswalks, normalized explicit aliases, target territory, and source measure. Code-only matches between distinct profiles—for example Tetun Prasa and Tetun Terik—are routed to review and never accepted automatically. Parent/child alternatives, bilingual categories, script variants, household measures, oral ability, and proficiency measures retain nonadditive overlap groups. Every final Top 10 and Top 100 entry requires a rankable exact production target and a registered person observation copied from a dated authority or calculated by a fully exposed derivation from dated inputs. The population measure need not equal the production audience: where a census label pools standards, communities, or orthographies, the count is retained once as a ceiling and the mismatch is stated. Unresolved macro-targets appear in a separate research appendix.

6.2 Factor assignment

For a missing exact natural-language FR-2 edition, $D=1$. For an exact named surface, $C=1$ is a target-definition identity conditional on literacy in that standard—not a claim that every speaker understands mathematics. Language-specific literacy is used where reported; otherwise a national literacy value appears only in a wide proxy sensitivity with low 0 and high 1. Internet and electricity select delivery modes rather than comprehension.

Non-overlap is factored into existing-edition, academic-lingua-franca, and selected- portfolio residuals. India C-17 supports a disclosed sensitivity (Office of the Registrar General & Census Commissioner, India, 2018): the low scenario treats every reported English-or-Hindi speaker as already served, the base treats half as comfortably served, and the high treats one fifth as served. Elsewhere, unknown academic overlap is [0, 1]; a value of .5 appears only in a neutral sensitivity, not as observed evidence.

The implemented paper consequently reports five views: exact gross ceiling per common compute; the direct-evidence marginal floor; base and optimistic factor-model sensitivities; a scarcity-adjusted sensitivity; and fail-closed interlanguage architecture comparisons. Vitality/equity, prestige, feasibility, and dialect risk remain descriptive or eligibility evidence; the frozen selector does not manufacture numeric portfolio objectives for them.

6.3 Compute scenarios

The workload ledger separately counts uncached input, cached input, and output. It does not recover or model cache-write tokens as a separate category; the API-equivalent dollar scenarios therefore exclude any cache-write surcharge. Gross, fresh, cached, weekly-plan, and monetary quantities are not interchangeable. For FR-2 D3, the low/base/high scenarios are:

Compute comparator. One exact FR-2/D3 edition under the three planning workflows.

Scenario	Gross tokens	API-equivalent USD	Status
Low	912,737	5.19	Planning workflow
Base	4,073,049	16.77	Planning workflow
High	18,664,571	65.43	Stress workflow

These dated dollar values are API-equivalent comparisons under the recorded GPT-5.6 Sol prices—$4 per million uncached input tokens, $0.40 per million cached input tokens, and $20 per million output tokens—accessed August 25, 2026 (OpenAI, 2026). They are not measured weekly Codex-plan consumption. The base scenario gives every unit an independent critique and allows substantial correction; the high scenario adds multiple broad passes and retry allowance. The full 722-unit Open Logic corpus costs 2.679/11.666/53.200 million gross tokens in the same low/base/high model.

The Indonesian remainder is separately estimated at 1.482/6.652/30.716 million gross tokens. No category-resolved usage ledger was recovered; all scenario coefficients remain explicit extrapolations.

6.4 Portfolio selection and double counting

The selector operates on canonical intervention cells rather than adding every available language count. Upstream overlap groups choose one observation when a parent category and same-name component are alternative measures. A shared-core package can replace enumerated natural-edition rows only if the validated population-link and intervention-matrix layers explicitly authorize their union. The current frozen interlanguage model authorizes none: all raw package links remain nonadditive and all candidate matrix rows remain nonrankable. The headline portfolio therefore retains the natural-edition rows and reports package aggregation separately as an engineering sensitivity. Multilingual speaker totals, multiple-response language counts, pooled script or orthography audiences, and accessibility strata retain explicit nonadditivity rules. No unobserved cross-language audience is subtracted or added dynamically.

The conservative efficiency view is reported even when every candidate has a zero floor. Because that view is then a complete tie and supplies no ordering information, admission cycles deterministically through three informative views: base marginal access per million base tokens, optimistic access per million low-scenario tokens, and scarcity-adjusted base access per million base tokens. Ties receive rank intervals. The cycle repeats until every eligible intervention is exposed; the first 10 and first 100 positions form the headline portfolios. Accessibility mechanisms remain in a separate safeguard backlog until a residual population gain can be bounded, rather than filling linguistic positions with global or nested ceilings.

6.5 Interlanguage compute comparison

A constructed surface receives no demographic reach merely from a family name. It can enter a future cardinal comparison only after an exact target/task comprehension gain is registered; under the present conservative rule, constructed Interslavic therefore has zero cross-language reach. A shared semantic core is different: it can produce a fully named native-language or script-localized output for every constituent, but shared-source reuse changes compute rather than readership. The analysis therefore compares complete named-output bundles with their independent-production counterparts while preserving each constituent population cell separately. Arithmetic reconciliation of those cells is necessary but not sufficient for a cardinal union; the frozen upstream nonadditivity controls keep every bundle outside the headline Top 10 and Top 100. The paper varies the adaptation fraction widely because no complete category-resolved local usage ledger establishes one universal reuse rate.

7. Results

7.1 Which populations are most affected?

The machine-readable Table 2 reports all 133 canonical natural-language interventions as source-bounded gross ceilings; the printed excerpt shows the ten largest base ceilings. These values retain each source's territory, year, age universe, question, and measure and are not interchangeable estimates of people harmed by language mismatch.

Table 2. Ten largest source-bounded gross population ceilings in the eligible candidate set.

Display order	Intervention	Profile(s)	Gross base ceiling	Measure	Year(s)	Source ID(s)
1	Bangladesh Bangla	bn-Beng-BD	165,323,060	derived_cldr_territory_functional_language_users	2026-03-17 CLDR release; component estimate year not stated	ASSEC-S006
2	Indian Bengali	bn-Beng-IN	96,177,835	census_same_name_mother_tongue_component_persons	2011	PM-S001
3	Vietnamese	vi-Latn-VN	89,000,000	peer_reviewed_published_lower_bound_speakers	2016 publication; population estimate vintage not stated	ASSEC-S007
4	Western Punjabi	pnb-Arab-PK	88,915,544	census_mother_tongue_persons	2023	PM-S003
5	Marathi	mr-Deva-IN	82,801,140	census_same_name_mother_tongue_component_persons	2011	PM-S001
6	Telugu	te-Telu-IN	80,912,459	census_same_name_mother_tongue_component_persons	2011	PM-S001
7	Indian Tamil	ta-Taml-IN	68,888,839	census_same_name_mother_tongue_component_persons	2011	PM-S001
8	Javanese	jv-Latn-ID	68,044,660	daily_language_at_home_age5plus_persons	2010	PM-S005
9	Thai	th-Thai-TH	64,080,191	derived_union_of_mutually_exclusive_census_strata_persons	2010-09-01	TH-S001

Display order	Intervention	Profile(s)	Gross base ceiling	Measure	Year(s)	Source ID(s)
10	Gujarati	gu-Gujr-IN	55,036,204	census_same_name_mother_tongue_component_persons	2011	PM-S001

Note. Population IDs retain their registered universe, date, interval status, and caveats. Exact authorities, locators, witness hashes, and draft APA references are in `source_id_reference_crosswalk.csv`. Profile-authority IDs establish a language, script, orthography, or institutional standard only; they do not establish audience size, educational deficit, mutual intelligibility, or translation quality.

The gross ceilings cannot answer how many readers newly gain comfortable access. The complete machine-readable Table 3 keeps low, base, and high marginal-access values separate; the printed excerpt reports the headline Top 10. The base and optimistic columns are factor-model sensitivities, not observed harmed-population counts. All 133 low values are zero because at least one access or non-overlap factor lacks a positive direct lower bound. The conservative efficiency view is consequently degenerate: every intervention ties on the interval [1, 133], and that view is reported but not used to fill portfolio positions.

Table 3. Marginal-access ranges for the headline Top 10.

Position	Intervention	Direct lower bound	Base sensitivity	Optimistic ceiling	Factor basis
1	Indian Bengali	0	69,699,712	93,375,817	Territory literacy proxy; India C-17 overlap proxy
2	Bangladesh Bangla	0	65,302,609	165,323,060	Territory literacy proxy; unmeasured academic overlap
3	Telugu	0	57,833,869	78,144,198	Territory literacy proxy; India C-17 overlap proxy
4	Marathi	0	50,316,309	75,430,264	Territory literacy proxy; India C-17 overlap proxy
5	Vietnamese	0	42,777,849	89,000,000	Territory literacy proxy; unmeasured academic overlap
6	Javanese	0	32,661,437	68,044,660	Territory literacy proxy; unmeasured academic overlap
7	Indian Tamil	0	48,387,709	66,095,914	Territory literacy proxy; India C-17 overlap proxy
8	Western Punjabi	0	26,167,845	88,915,544	Territory literacy proxy; unmeasured academic overlap
9	Gujarati	0	34,089,878	50,467,358	Territory literacy proxy; India C-17 overlap proxy

Position	Intervention	Direct lower bound	Base sensitivity	Optimistic ceiling	Factor basis
10	Kannada	0	31,535,081	42,241,972	Territory literacy proxy; India C-17 overlap proxy

The evidence-ready gap map is genuinely global but sharply Asia-heavy: 57 of the 133 eligible exact interventions are in Asia (26 South Asian, 25 Southeast Asian, five Central Asian, and one East Asian); 24 are in Africa; 35 in the Americas, including Arctic, Indigenous, Caribbean/Central American diaspora, and Pacific-diaspora strata; 12 in Oceania; and five in Europe. The Top 100 contains all 57 Asian and all 24 African rows, 12 American rows, five European rows, and two Oceanian rows. These are counts of source-ready intervention records, not regional population totals. They show why a search starting only from small or endangered languages would miss large official languages whose readers may still lack comfortable open advanced mathematics, while a population-only list would miss Indigenous, signed, diaspora, and prestige-domain access. Appendix E preserves the exact subregions, source types, measures, and confidence counts without summing incompatible population universes.

7.2 Top 10 interventions

The selector alternates base, optimistic, and scarcity sensitivity lanes. Under the fail-closed rule, no shared-core package substitutes for its constituent natural-language rows. This is a portfolio order, not a claim of a uniquely identified welfare ranking.

Table 4. Headline Top-10 commissioning portfolio.

Position	Intervention	Profile(s)	Gross base ceiling	Base marginal sensitivity	Base gross tokens	Informative rank range	Next OpenStax allocation
1	Indian Bengali	bn-Beng-IN	96,177,835	69,699,712	4,073,049	1–9	SB-1 (D3)
2	Bangladesh Bangla	bn-Beng-BD	165,323,060	65,302,609	4,073,049	1–11	MV-1 (D3)
3	Telugu	te-Telu-IN	80,912,459	57,833,869	4,073,049	1–5	SB-1 (D3)
4	Marathi	mr-Deva-IN	82,801,140	50,316,309	4,073,049	2–6	SB-1 (D3)

Position	Intervention	Profile(s)	Gross base ceiling	Base marginal sensitivity	Base gross tokens	Informative rank range	Next OpenStax allocation
5	Vietnamese	vi-Latn-VN	89,000,000	42,777,849	4,073,049	3–17	SB-1 (D3)
6	Javanese	jv-Latn-ID	68,044,660	32,661,437	4,073,049	3–8	SB-1 (D3)
7	Indian Tamil	ta-Taml-IN	68,888,839	48,387,709	4,073,049	5–8	SB-1 (D3)
8	Western Punjabi	pnb-Arab-PK	88,915,544	26,167,845	4,073,049	4–12	MV-1 (D2)
9	Gujarati	gu-Gujr-IN	55,036,204	34,089,878	4,073,049	4–10	SB-1 (D3)
10	Kannada	kn-Knda-IN	43,506,272	31,535,081	4,073,049	9–13	SB-1 (D3)

Note. Gross populations retain their source measures; base access and informative rank ranges are model sensitivities. Full source IDs and profile authorities appear in Appendix B and the source crosswalk.

7.3 Top 100 portfolio

The numbered Top 100 contains 100 natural-language interventions and 100 separately named localized outputs; no package occupies a cardinal position. Accessibility remains a separate, non-cardinal safeguard backlog and does not consume a linguistic position. The frozen FR-2 workload sums to 91,273,700 low, 407,304,900 base, and 1,866,457,100 high gross tokens. These compute totals are additive; the heterogeneous population measures are not summed as a count of unique people.

Table 5. Top-100 reporting-category composition and base FR-2 compute.

Reporting category	Interventions	Localized outputs	Base / optimistic / scarcity admissions	FR-2 base gross tokens
high_reach_underserved	34	34	12 / 12 / 10	138,483,666
regional_depth	49	49	21 / 21 / 7	199,579,401
small_population_prestige_domain_screen	17	17	7 / 7 / 3	69,241,833

Note. Reporting categories are descriptive labels applied to selected rows, not additional admission objectives. The admission lanes are base, optimistic, and scarcity.

7.4 Interlanguage and shared-core comparisons

No package replaces a natural-language row in the fail-closed cardinal portfolio. IL-PUNJABI, IL-HU, IL-NGUNI, and IL-SOTHO retain complete component joins and positive modeled savings, but their upstream population unions are not admissible. Their displayed population values are diagnostic component subtotals only: they are nonadditive, noncardinal, and not unique-person, family-level, or constructed-bridge comprehension reach. Base reuse is an unobserved sensitivity, not an empirical production saving.

For the South African architectures, national curriculum and examination guidance supports distinct named outputs and distinguishes conjunctively written isiXhosa, isiNdebele, isiZulu, and siSwati from disjunctively written Sepedi, Sesotho, Setswana, Tshivenda, and Xitsonga (Department of Basic Education, n.d., 2015). That evidence supports output identity and orthographic workflow; it does not establish additive Nguni or Sotho-Tswana audiences or cross-language comprehension.

Table 6A. Base-scenario shared-core architecture sensitivities.

Architecture	Named outputs	Profiles	Diagnostic component subtotal (noncardinal)	Independent base tokens	Shared-core base sensitivity	Modeled saving sensitivity
IL-HU	2	ur-Arab-PK;ur-Arab-IN	72,975,069	8,146,098	6,109,574	2,036,524
IL-NGUNI	4	zu-Latn-ZA;xh-Latn-ZA;ss-Latn-ZA;nr-Latn-ZA	27,137,226	16,292,196	10,182,623	6,109,573
IL-PUNJABI	2	pa-Guru-IN;pnb-Arab-PK	120,059,639	8,146,098	6,109,574	2,036,524
IL-SOTHO	3	nso-Latn-ZA;tn-Latn-ZA;st-Latn-ZA	15,624,006	12,219,147	8,146,098	4,073,049

Table 6B. Full low/base/high shared-core token-reuse sensitivities.

Architecture	Workflow / reuse case	Outputs	Independent gross tokens	Shared-core gross tokens	Modeled saving	Saving fraction
IL-HU	Low / optimistic reuse	2	1,825,474	1,095,284	730,190	40.0%

Architecture	Workflow / reuse case	Outputs	Independent gross tokens	Shared-core gross tokens	Modeled saving	Saving fraction
IL-HU	Base / base reuse	2	8,146,098	6,109,574	2,036,524	25.0%
IL-HU	High / conservative reuse	2	37,329,142	33,596,228	3,732,914	10.0%
IL-NGUNI	Low / optimistic reuse	4	3,650,948	1,460,379	2,190,569	60.0%
IL-NGUNI	Base / base reuse	4	16,292,196	10,182,623	6,109,573	37.5%
IL-NGUNI	High / conservative reuse	4	74,658,284	63,459,541	11,198,743	15.0%
IL-PUNJABI	Low / optimistic reuse	2	1,825,474	1,095,284	730,190	40.0%
IL-PUNJABI	Base / base reuse	2	8,146,098	6,109,574	2,036,524	25.0%
IL-PUNJABI	High / conservative reuse	2	37,329,142	33,596,228	3,732,914	10.0%
IL-SOTHO	Low / optimistic reuse	3	2,738,211	1,277,832	1,460,379	53.3%
IL-SOTHO	Base / base reuse	3	12,219,147	8,146,098	4,073,049	33.3%
IL-SOTHO	High / conservative reuse	3	55,993,713	48,527,885	7,465,828	13.3%

Note. Reuse fractions are unobserved engineering assumptions, not recovered production savings. All twelve rows remain noncardinal because both frozen upstream union controls are false. Exact arithmetic and source identities are preserved in `SHARED_CORE_ARCHITECTURE_LOW_BASE_HIGH.csv`.

The broader overlap register contains 80 interlanguage rows across 15 intervention IDs. It separates constructed surfaces, dual-script/register production, pluricentric shared cores, and natural-intercomprehension reuse. Arabic, BCMS, Hindi-Urdu, Indonesian/Malay, Manding, Persian-Dari-Tajik, Punjabi, Nguni, Sotho-Tswana, Southern Quechua, Romance, Scandinavian, Turkic, Germanic, and Interslavic hypotheses are all represented, but no component is cardinally rankable under current evidence.

The proposed Inter-Germanic layer is therefore an architecture hypothesis, not a single new audience. Its exact candidate communities are Afrikaans, Dutch, Faroese, Hutterisch, Low German, Pennsylvania Dutch, Plautdietsch, Scots, West Frisian, and a defined Yiddish register. English, German, Russian, Hebrew, Danish, and other existing academic-language overlaps vary by community, and no target/task study supports a pan-Germanic comprehension multiplier. A defensible program would reuse terminology and source structure across separately named West/North/diaspora outputs; it would not release one pan-Germanic text and sum those populations. Appendix H gives the exact community, script, prior-study, overlap, exclusion, and double-count rule for every registered interlanguage row.

The other 11 package proposals are excluded because planned named-output coverage, exact population joins, constituent score joins, or interval reconciliation is incomplete. Constructed Interslavic is assigned zero conservative cross-language reach because no exact target/task incremental-comprehension result is registered; no language-family population is credited to that surface. Reuse fractions for every architecture remain modeled sensitivities rather than observed production savings.

7.5 OpenStax and Open Logic curriculum allocation

FR-2 at D3 (210 editable units; 120,083 measured source alpha tokens) is the common first product. Among the 100 natural-language Top-100 rows, the next-step allocation is 36 to MV-1 at D2, 44 to MV-1 at D3, and 20 to SB-1 at D3. Each row is one named output, so these groups also cover 36, 44, and 20 outputs, respectively.

Table 7. Next OpenStax allocations for the 100 cardinal editions.

Next portfolio	Depth	Interventions	Outputs	Low tokens	Base tokens	High tokens
MV-1	D2	36	36	39,137,220	156,816,432	670,681,044
MV-1	D3	44	44	70,946,876	240,856,792	927,247,244
SB-1	D3	20	20	59,947,340	227,482,500	942,247,140

Note. MV-1 is the full Algebra and Trigonometry 2e collection; SB-1 adds Introductory Statistics 2e. Measured and planning-interval source denominators are kept distinct in Appendix F and the machine-readable allocation table.

Table 7 costs only the 100 cardinal natural-language interventions. Architecture sensitivities remain in Table 6 and are not transferred into OpenStax compute or cardinal output totals.

7.6 Sensitivity and negative controls

The informative rank range spans the base, optimistic, and scarcity views. The conservative zero floor is disclosed separately and does not manufacture an arbitrary strict order. Richly served controls remain available for non-overlap, subgroup, and accessibility tests but are not generic translation priorities. Unresolved profiles and D0 profile/population mismatches are exclusions, not evidence that their communities are already served.

Across every linguistic commission, semantic HTML with structured accessible mathematics is the foundational safeguard rather than a later language competitor. Its FR-2 base increment is 610,957 gross tokens in the registered scenario. Formula narration, proof-tree and figure remediation, tagged PDF, offline packaging, plain-language companions, Braille, and exact signed-language products follow as separate derivatives with overlapping beneficiary ceilings. Appendix C gives their ordered specifications and costs; only the first item's foundational priority is evidence-backed, while positions 2-11 preserve deterministic backlog order rather than claiming a cardinal welfare rank.

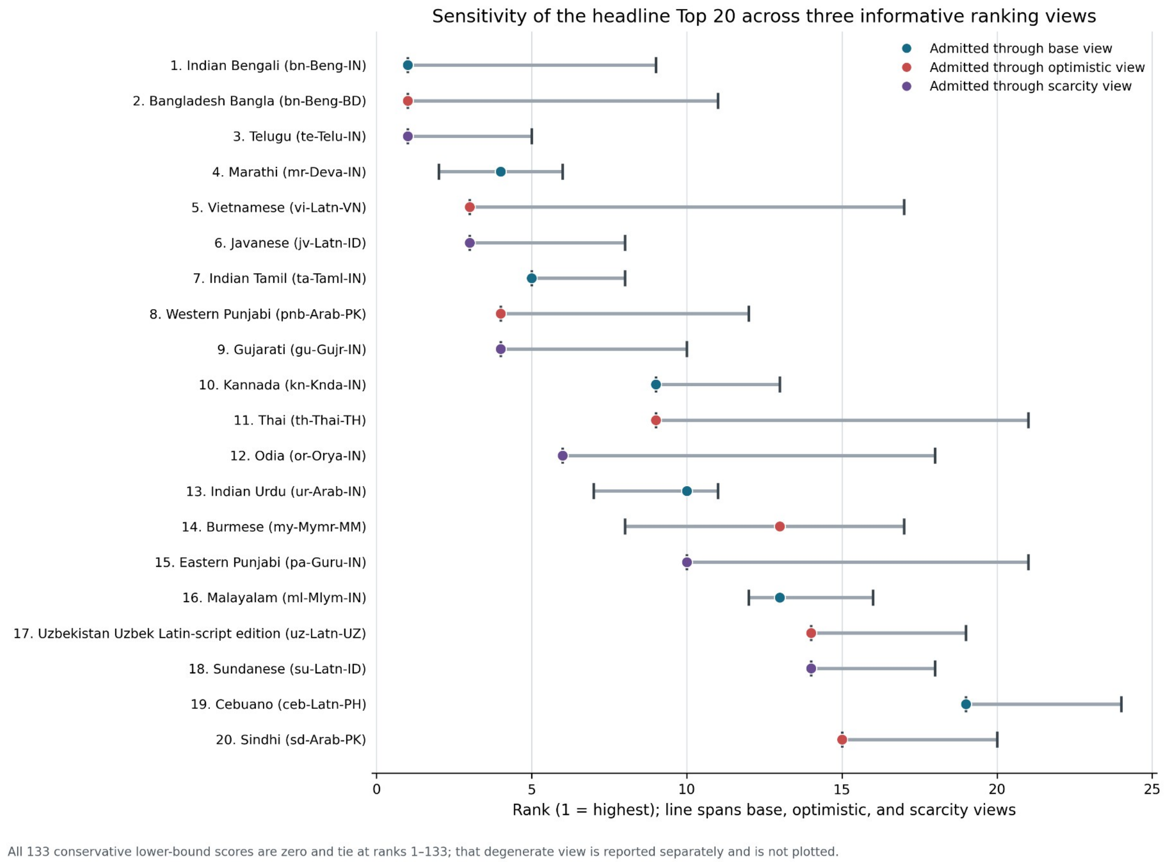

Figure 1. Sensitivity of the headline Top 20 across the three informative ranking views. All 133 direct-evidence lower bounds are zero and tie at ranks 1–133, so that degenerate conservative view is disclosed but not plotted.

Table 8. Richly served negative controls retained outside the portfolio.

Target	Profile	Territory	Gross base population	Measure	Year	Source ID	Classification
Polish	pl-Latn-PL	Poland	37,868,618	census_languages_used_at_home_multiple_response_persons	2021-03-31	GRG-S006	richly_served_negative_control
German	de-Latn	Poland	216,342	census_languages_used_at_home_multiple_response_persons	2021-03-31	GRG-S006	richly_served_negative_control

Target	Profile	Territory	Gross base population	Measure	Year	Source ID	Classification
Ukrainian	uk-Cyrl-UA	Poland	55,104	census_languages_used_at_home_multiple_response_persons	2021-03-31	GRG-S006	richly_served_negative_control
Czech	cs-Latn-CZ	Poland	5,328	census_languages_used_at_home_multiple_response_persons	2021-03-31	GRG-S006	richly_served_negative_control

Note. These multiple-response home-language counts are not global language populations. They test whether a generic full-edition recommendation survives when the target is already richly served; subgroup and accessibility work may still be valuable.

Table 8 also preserves 5 non-control exclusions: three unresolved localized-output profiles and two D0 profile/population mismatches. They are not counted as negative controls and must not be silently converted into recommendations.

The complete machine-readable tables preserve exact population measures, years, observation IDs, source IDs, factor statuses, compute scenarios, package exclusions, and tie-aware ranks.

8. Grant and implementation program

The recommended program is staged by reusable infrastructure rather than by isolated PDFs.

1. **Core source pipeline.** Pin editable Open Logic/OpenStax units, formula identity, licensing, semantic structure, tests, and deterministic builds.
2. **High-reach wave.** Produce exact-language FR-2 editions for the highest stable marginal-access-per-compute targets, followed by the recommended OpenStax gateway.
3. **Shared-core wave.** Where evidence supports pluricentric or related-language reuse, build one semantic/terminology core but release explicitly named localized outputs.
4. **Regional-depth and prestige wave.** Fund smaller exact profiles whose scarcity, vitality, regional function, or serious-mathematics prestige value would be erased by population ranking.

5. **Accessibility wave.** Derive HTML/MathML, reflowable EPUB, accessible PDF, offline, plain-language, audio/Braille, and exact national signed-language products according to the relevant barrier and population stratum.
6. **Maintenance and evaluation.** Preserve source hashes, terminology decisions, formula/structure checks, build evidence, model-audit results, corrections, and privacy-preserving usage indicators. New comprehension or usage evidence updates the next portfolio without making human availability a release prerequisite.

Table 9. Grant-scale low/base/high inference scenarios.

Scope	Workflow	Cardinal interventions	Modeled outputs	Task bundle	Gross tokens	API-equivalent USD
Top-10 pilot	Low	10	10	10 x FR-2/D3	9,127,370	$51.95
Top-10 pilot	Base	10	10	10 x FR-2/D3	40,730,490	$167.72
Top-10 pilot	High	10	10	10 x FR-2/D3	186,645,710	$654.29
Headline Top 100	Low	100	100	100 x FR-2/D3	91,273,700	$519.50
Headline Top 100	Base	100	100	100 x FR-2/D3	407,304,900	$1,677.21
Headline Top 100	High	100	100	100 x FR-2/D3	1,866,457,100	$6,542.92
Top 100 plus next OpenStax allocation	Low	100	200	FR-2 plus 36 x MV-1/D2, 44 x MV-1/D3, 20 x SB-1/D3	261,305,136	$1,952.39
Top 100 plus next OpenStax allocation	Base	100	200	FR-2 plus 36 x MV-1/D2, 44 x MV-1/D3, 20 x SB-1/D3	1,032,460,624	$5,935.91
Top 100 plus next OpenStax allocation	High	100	200	FR-2 plus 36 x MV-1/D2, 44 x MV-1/D3, 20 x SB-1/D3	4,406,632,528	$22,220.90
All 133 eligible linguistic interventions	Low	133	136	136 x FR-2/D3	124,132,232	$706.51
All 133 eligible linguistic interventions	Base	133	136	136 x FR-2/D3	553,934,664	$2,281.00
All 133 eligible linguistic interventions	High	133	136	136 x FR-2/D3	2,538,381,656	$8,898.37

Note. The 133-row eligible inventory emits 136 full outputs because three source-audited intervention rows require two localized outputs while retaining one pooled, nonadditive population interval. Dollar values reprice aggregate uncached input, cached input, and output tokens at the dated API rates. Cache-write tokens were not observed and are neither assumed nor set to zero. Non-API program costs are excluded, not zero.

Budget requests should separate inference, edition, and delivery costs. A low API bill does not imply low all-in cost where source acquisition, terminology, typography, accessibility, teacher materials, or distribution dominate. Conversely, openly licensed semantic infrastructure creates option value across many future languages, editions, and revisions.

9. Limitations

Population sources differ in year, age universe, language question, and territorial coverage. Exact counts are exact only to their printed measure; they are not all current L1-speaker estimates. Many censuses measure ethnicity rather than language, speaking rather than reading, or household use rather than individual ability.

Academic-lingua-franca comfort is the largest unresolved factor. Official status, school exposure, or self-reported subsidiary-language knowledge does not prove comfortable advanced reading. Wide ranges are therefore a feature, not an error.

Scarcity proxies can miss unindexed, offline, paywalled, or locally hosted resources, and can overcount research works that are not teaching material. Model benchmarks are general-domain and may not represent specialist mathematics or the target variety.

The compute model is reproducible but scenario-based. It is calibrated to exact local source tokens and workflow stages, not to a recovered category-resolved token ledger. API-equivalent prices are date- and model-specific and should not be treated as weekly subscription accounting or complete program cost.

Interlanguage evidence is especially sparse. Short-task comprehension, linguistic relatedness, or a family label cannot establish sustained mathematical usability. The conservative portfolio accordingly awards zero cross-language reach where exact evidence is absent, while shared-core localized-output architectures remain eligible for compute savings.

Finally, access is not learning. The ranking identifies where a source-faithful open edition removes a barrier. Learning-adjusted claims require a defined instructional package, delivery, exposure, and outcome evaluation.

10. Conclusion

Efficient use of AI compute for educational access is a portfolio-design problem, not a speaker-count sort and not a race to maximize translated file totals. The central unit must be an exact production profile—language or variety, script, orthography, territory, register, and curriculum—joined to a dated population observation and an explicit non-overlap model. That requirement changes both who appears near the top and how uncertainty is communicated. Bangladesh Bangla has the largest gross ceiling in the eligible candidate set, but Indian Bengali enters first under the base sensitivity; Telugu enters through scarcity; Vietnamese and Western Punjabi enter through the optimistic view. Their order records the chosen portfolio rule rather than pretending that one uncertain scalar identifies global welfare.

The practical first wave is therefore the frozen Top 10, each beginning with the same 210-unit Formal Reasoning Core so that language priorities remain content-comparable. The non-padded Top 100 then expands global and regional depth; after selection, 17 entries fall in the small-population/prestige-domain reporting category. Its 100 exact editions cost an estimated 91.274 million gross tokens in the low workflow, 407.305 million in the base workflow, and 1.866 billion in the high workflow. The next curriculum layer allocates 36 editions to the MV-1 D2 gateway, 44 to MV-1 D3, and 20 to the SB-1 D3 statistics/data gateway. These are reproducible inference-workload scenarios, not complete program budgets; source acquisition, typography, accessibility, distribution, and maintenance remain real costs.

Interlanguage work is most defensible here as production architecture. Punjabi, dual-territory Urdu, Nguni, and Sotho shared-core hypotheses could reduce modeled base compute for their named outputs by 25%–37.5%, but those savings are unobserved sensitivities and their upstream population unions remain nonadditive. They therefore do not displace natural-language rows in the Top 100. Constructed Interslavic similarly receives no family demographic multiplier from a short cloze result. This is not a rejection of interlanguage design: it identifies the exact evidence or production ledger needed for one to outperform independently named editions without double counting readers.

Finally, written-language count is not a complete access measure. The separate 11-intervention safeguard backlog preserves MathML/HTML, reflowable, offline, plain-language, audio/Braille, and signed-language needs without falsely summing nested global disability or connectivity ceilings. The recommended program is thus to fund a reusable semantic and formula-preserving source pipeline, execute the Top 10 and then the diversified Top 100, test shared-core reuse as a separately instrumented workflow, and generate accessibility derivatives alongside—not after—the linguistic editions. As better literacy, academic-comfort, comprehension, production, and usage evidence arrives, the same open model can be rerun without discarding prior editions or hiding the uncertainty that motivated the portfolio in the first place.

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Appendix A. Reproducibility and computational provenance

The research synthesis, source extraction, deterministic modeling, code, and manuscript production were performed with **OpenAI Codex gpt-5.6-sol, Ultra**. This identification does not replace the named authorities in the source register. Exact source labels, measures, locators, dates, caveats, and local witness hashes remain attached to the corresponding machine-readable rows.

The table below identifies the principal frozen inputs and outputs used by the paper. `MANIFEST.sha256` supplies the complete public-file inventory at release time.

Artifact	Data rows	Bytes	SHA-256
population_observations_master.csv	474	386,495	2BD6D9F73D746C5FEF18D05CECC7E397DA2A9DD4C38 EA33EEE15A1BBC75D5BD9
population_source_register_master.csv	80	69,063	B3FFF7785829ABF868FD0A05D8D883F5AE83AD44B0CC EFAA4898FA0F3E3E184F
MODEL_SPEC.md	—	8,522	0D7254C61152DCB1207AA2AA0EEC1DDF2599EFB23631 C9C2B6276005BB4C96ED
RANKING_METHOD.md	—	3,375	AB7FB1ECBC1728CB649C024AE3A49542166726CF7362 8117B2735EA8F01381BC
FACTOR_POLICY.md	—	9,754	36724C1D3071A2578FEA5C0015233627952210085CC05 C39EC6E236732BC60A9
candidate_interventions_master.csv	209	122,251	DA89EBE46700D21820579774B75C39D79C224C2FF626 76C3243CA16D13DCBC21
natural_language_scores_v3.csv	105	137,228	02BAB54BC4714854003265F2A42BCC87BC400C99650D A90764192BC00A87BD4B
candidate_expansion_scores.csv	37	50,444	B97383E20E9FAD2791BD4883082CF3985C66F08158670 A466B9D82597BD43291
staging/interlanguage_bundle_model/interlanguage_intervention_matrix.csv	104	84,768	F3F9BA46524BBF4754E2F12CFEC2AD5D6A24377B786C 8BFE1AF3EE6AD819889F
staging/interlanguage_bundle_model/interlanguage_population_links.csv	113	49,082	8743D37ACB11EC2942EB3BE07AC9C4F884FD07DA8FCF F320E89B6775EC5A6DEB

Artifact	Data rows	Bytes	SHA-256
interlanguage_package_scores.csv	135	88,114	5B943C3CDF2C2454F21ADD46B8897F1F61CFE32F06CA81E1561378D8C855BAE2
portfolio_compute_scenarios.csv	60	18,067	6FCFBE82135F0671BF30F03FE1F404FF8E9E9C054992A4445AA8ACA0016A4961
curriculum_units.csv	29	7,074	0F7A958D7C19F989A1C95F673456B3667ED23F41B64208100EA1CB5C3360E204
curriculum_portfolios.csv	13	3,404	2E4CC460C4CB343A092C5A8A4C11EC09EEF091A93807D983B74E258A698B5049
openstax_workload_measurements.csv	94	26,411	ECF9E83428CFCBA41600AE070F970A9685E2C5F7E6E2F2045C07B1DB319A86A2
portfolio_linguistic_candidates.csv	133	163,766	BB1EB6688E0FA1EA7E902C46BCDC4EA55C8ED83F4334A5AE97F0E9BA83BC58B0
TOP_10.csv	10	12,498	C6519943F525881633603D28E43C0F32090ABEA2F560BDE5B117C755C95C4060
TOP_100.csv	100	117,831	C2AE2043DA6A136055C439B39533CCAB3B568E27BC90E5B9B2B8A538C00867AE
accessibility_priority_backlog.csv	11	12,375	3FCD6060C48A44618B157CC503839921E0DEA443A16ACF06B51E1C3D8D21B46F
figure_1_rank_sensitivity_data.csv	20	1,875	5DD1818380DB1DA6E7F1175D08D4D3E140748EE9B2F29F0031714ED9728425AA
SHARED_CORE_ARCHITECTURE_LOW_BASE_HIGH.csv	12	4,270	C54F4497C0236F80071198E30BFEB3913A520245EEBE885D17626F911930B64F
GRANT_COMPUTE_SCENARIOS_LOW_BASE_HIGH.csv	12	9,074	5EC3B6277A7D0CADC9FA26D62E5B13F0926752640255C93B62AB409C2ADE02C8
IMPLEMENTED_VS_UNIMPLEMENTED_SENSITIVITY_MATRIX.csv	26	4,912	9573E4FD72DD03E1827F1D7EEB5F1EE8FC42172B03BBF68423DEEE62975E13D1
appendix_a_existing_work_reconciliation.csv	209	113,976	0679AA3B81557E4CFBFAD6F8C84EEFC6D4898D13CE576B5C15B0D5808DA6CFE8
appendix_b_global_gap_map.csv	133	256,403	2EBB1EACE0BDBFBF4BE7BCF7F294AE2F19D9C6B35870DD40C99DDCF4302CF46F
appendix_c_accessibility_safeguards.csv	11	14,626	87575CD943D6D76FBAC866A9D1CB4A9BEC1B5AB8EF5B6B6C2131282256FC71F9

Artifact	Data rows	Bytes	SHA-256
appendix_d_top100_curriculum_mapping.csv	100	47,197	C6E088B66CAF7C332284F1733BCB9F895F53C1A82E5D1AFEDD2F4FA9B52DD371
appendix_e_unresolved_profiles_and_d0.csv	5	5,026	5B542465C2AF03CD9CAA6127E2E45024EC4A944FAAEE6738275F76C6C8AB306A
appendix_f_interlanguage_matrix_summary.csv	15	19,470	B5E48C2928F7D8B85152DF721A9DEC588ED3FF130D63D04689DBCDB9724E6158
source_id_reference_crosswalk.csv	54	100,111	81173631459A66D413B067407D0C34C899FF64654E4C2B6C254F05F8EB64ECB2
corpus_method_reference_crosswalk.csv	21	31,848	698B907E2323B3A96E90EC8099EE1B2C1C471EEFBAD4B8FC106CEAFD4D2CE2A9

Appendix B. Full ordered Top 100

Every row is one exact named natural-language edition. Population values retain the source's own measure and date; they are not summed as unique people. `Base access` is a labelled sensitivity, not an observed harmed-population count.

Population and profile source IDs resolve through `source_id_reference_crosswalk.csv`. Fifty-one of the 54 used source IDs have hash-verified local witnesses; PM-S005, PM-S014, and PM-S016 retain complete registered metadata but no registered local witness and are explicitly marked `NO_WITNESS_REGISTERED` in that crosswalk.

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
1	Indian Bengali	bn-Beng-IN	India	96,177,835	69,699,712	high_reach_underserved	SB-1 (D3)	PM-S001
2	Bangladesh Bangla	bn-Beng-BD	Bangladesh	165,323,060	65,302,609	high_reach_underserved	MV-1 (D3)	ASSEC-S006
3	Telugu	te-Telu-IN	India	80,912,459	57,833,869	high_reach_underserved	SB-1 (D3)	PM-S001
4	Marathi	mr-Deva-IN	India	82,801,140	50,316,309	high_reach_underserved	SB-1 (D3)	PM-S001
5	Vietnamese	vi-Latn-VN	Vietnam	89,000,000	42,777,849	high_reach_underserved	SB-1 (D3)	ASSEC-S007
6	Javanese	jv-Latn-ID	Indonesia	68,044,660	32,661,437	high_reach_underserved	SB-1 (D3)	PM-S005
7	Indian Tamil	ta-Taml-IN	India	68,888,839	48,387,709	high_reach_underserved	SB-1 (D3)	PM-S001
8	Western Punjabi	pnb-Arab-PK	Pakistan	88,915,544	26,167,845	high_reach_underserved	MV-1 (D2)	PM-S003
9	Gujarati	gu-Gujr-IN	India	55,036,204	34,089,878	high_reach_underserved	SB-1 (D3)	PM-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
10	Kannada	kn-Knda-IN	India	43,506,272	31,535,081	high_reach_underserved	SB-1 (D3)	PM-S001
11	Thai	th-Thai-TH	Thailand	64,080,191	29,188,527	high_reach_underserved	SB-1 (D3)	TH-S001
12	Odia	or-Orya-IN	India	34,059,266	23,106,806	high_reach_underserved	SB-1 (D3)	PM-S001
13	Indian Urdu	ur-Arab-IN	India	50,725,762	30,600,888	high_reach_underserved	SB-1 (D3)	PM-S001
14	Burmese	my-Mymr-MM	Myanmar	36,817,344	17,217,151	high_reach_underserved	MV-1 (D3)	ASSEC-S006
15	Eastern Punjabi	pa-Guru-IN	India	31,144,095	17,886,376	high_reach_underserved	SB-1 (D3)	PM-S001
16	Malayalam	ml-Mlym-IN	India	34,776,533	23,946,869	high_reach_underserved	SB-1 (D3)	PM-S001
17	Uzbekistan Uzbek Latin-script edition	uz-Latn-UZ	Uzbekistan	35,700,000	17,850,000	high_reach_underserved	SB-1 (D3)	ALG-S018
18	Sundanese	su-Latn-ID	Indonesia	32,412,752	15,558,121	high_reach_underserved	SB-1 (D3)	PM-S005
19	Cebuano	ceb-Latn-PH	Philippines	20,697,364	10,190,347	high_reach_underserved	SB-1 (D3)	ASSEC-S008
20	Sindhi	sd-Arab-PK	Pakistan	34,401,564	10,124,380	high_reach_underserved	MV-1 (D2)	PM-S003
21	Amharic	am-Ethi-ET	Ethiopia	21,634,396	6,540,078	high_reach_underserved	MV-1 (D2)	PM-S020
22	Assamese	as-Beng-IN	India	14,816,414	10,118,381	high_reach_underserved	SB-1 (D3)	PM-S001
23	Dari	prs-Arab-AF	Afghanistan	30,083,114	5,605,988	high_reach_underserved	MV-1 (D2)	ASSEC-S018
24	Pakistani Urdu	ur-Arab-PK	Pakistan	22,249,307	6,547,971	high_reach_underserved	MV-1 (D2)	PM-S003
25	Saraiki	skr-Arab-PK	Pakistan	28,849,579	8,490,431	high_reach_underserved	MV-1 (D2)	PM-S003

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
26	Afghanistan Pashto Arabic-script edition	ps-Arab-AF	Afghanistan	18,753,110	3,494,642	high_reach_underserved	MV-1 (D2)	ASSEC-S018
27	Sinhala	si-Sinh-LK	Sri Lanka	14,948,168	6,931,465	high_reach_underserved	MV-1 (D3)	ASSEC-S006
28	India Maithili Devanagari edition	mai-Deva-IN	India	13,353,347	7,644,770	high_reach_underserved	SB-1 (D3)	PM-S001
29	Khmer	km-Khmr-KH	Cambodia	14,893,134	5,356,316	high_reach_underserved	MV-1 (D3)	ALG-S011
30	Nepali	ne-Deva-NP	Nepal	13,084,457	5,502,014	high_reach_underserved	MV-1 (D2)	PM-S004
31	isiZulu	zu-Latn-ZA	South Africa	14,613,202	6,659,967	high_reach_underserved	MV-1 (D3)	ZA-S001
32	Kazakh	kk-Cyrl-KZ	Kazakhstan	13,380,107	3,345,027	high_reach_underserved	SB-1 (D3)	ALG-S007
33	Turkmen	tk-Latn-TM	Turkmenistan	6,297,965	3,145,834	regional_depth	MV-1 (D3)	ALG-S009
34	Shona	sn-Latn-ZW	Zimbabwe	11,251,734	5,244,996	high_reach_underserved	MV-1 (D2)	PM-S011
35	Tajik	tg-Cyrl-TJ	Tajikistan	10,394,100	2,598,525	high_reach_underserved	SB-1 (D3)	ASSEC-S006
36	Lao	lo-Laoo-LA	Laos	5,487,956	2,075,545	regional_depth	MV-1 (D2)	ASSEC-S006
37	isiXhosa	xh-Latn-ZA	South Africa	9,786,928	4,460,393	regional_depth	MV-1 (D3)	ZA-S001
38	Mali Bamanankan Latin-script edition	bm-Latn-ML	Mali	9,551,561	1,701,611	regional_depth	MV-1 (D2)	AFG-S008
39	Ethiopia Somali Latin-script edition	so-Latn-ET	Ethiopia	4,609,274	1,393,384	regional_depth	MV-1 (D2)	PM-S020
40	Indonesia Madurese Latin-script edition	mad-Latn-ID	Indonesia	7,743,533	3,716,896	regional_depth	MV-1 (D3)	PM-S005
41	Wolof	wo-Latn-SN	Senegal	8,786,892	2,212,539	regional_depth	MV-1 (D2)	AFG-S009

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
42	Kyrgyz	ky-Cyrl-KG	Kyrgyzstan	5,527,168	1,381,792	regional_depth	MV-1 (D3)	ALG-S008
43	Ilocano	ilo-Latn-PH	Philippines	7,098,503	3,494,948	regional_depth	MV-1 (D3)	ASSEC-S008
44	Hiligaynon	hil-Latn-PH	Philippines	6,073,883	2,990,476	regional_depth	MV-1 (D3)	ASSEC-S008
45	Sri Lankan Tamil	ta-Taml-LK	Sri Lanka	3,297,390	1,529,000	regional_depth	MV-1 (D3)	ASSEC-S006
46	South Africa Sepedi Latin-script edition	nso-Latn-ZA	South Africa	5,972,255	2,721,855	regional_depth	MV-1 (D3)	ZA-S001
47	South Africa Setswana Latin-script edition	tn-Latn-ZA	South Africa	4,972,787	2,266,348	regional_depth	MV-1 (D3)	ZA-S001
48	Mongolian	mn-Cyrl-MN	Mongolia	3,051,962	1,505,228	regional_depth	MV-1 (D3)	ASSEC-S006
49	South Africa Sesotho Latin-script edition	st-Latn-ZA	South Africa	4,678,964	2,132,438	regional_depth	MV-1 (D3)	ZA-S001
50	Ethiopia Tigrinya Ethiopic-script edition	ti-Ethi-ET	Ethiopia	4,324,933	1,307,427	regional_depth	MV-1 (D2)	PM-S020
51	Maori	mi-Latn-NZ	New Zealand	213,849	53,462	regional_depth	MV-1 (D3)	PM-S014
52	Indonesia Minangkabau Latin-script edition	min-Latn-ID	Indonesia	4,232,226	2,031,468	regional_depth	MV-1 (D3)	PM-S005
53	Senegal Pulaar, official Latin orthography	fuc-Latn-SN	Senegal	4,314,509	1,086,393	regional_depth	MV-1 (D2)	AFG-S009
54	Ireland Irish Latin-script daily-use edition	ga-Latn-IE	Ireland	71,968	17,992	small_population_prestige_domain_screen	MV-1 (D3)	PM-S016
55	Indonesia Banjar Latin-script edition	bjn-Latn-ID	Indonesia	3,651,626	1,752,780	regional_depth	MV-1 (D3)	PM-S005
56	Ethiopia Sidama Latin-script edition	sid-Latn-ET	Ethiopia	2,981,471	901,299	regional_depth	MV-1 (D2)	PM-S020
57	Official-standard Belarusian	be-Cyrl-BY	Poland	17,325	4,331	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S006
58	Waray	war-Latn-PH	Philippines	2,694,135	1,326,457	regional_depth	MV-1 (D3)	ASSEC-S008

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
59	South Africa Xitsonga Latin-script edition	ts-Latn-ZA	South Africa	2,784,279	1,268,935	regional_depth	MV-1 (D3)	ZA-S001
60	Standard Lithuanian	lt-Latn-LT	Poland	5,422	1,356	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S006
61	Indonesia Sasak Latin-script edition	sas-Latn-ID	Indonesia	2,691,127	1,291,741	regional_depth	MV-1 (D3)	PM-S005
62	Brahui	brh-Arab-PK	Pakistan	2,778,670	817,763	regional_depth	MV-1 (D2)	PM-S003
63	Indonesia Acehnese Latin-script edition	ace-Latn-ID	Indonesia	2,550,055	1,224,026	regional_depth	MV-1 (D3)	PM-S005
64	Kirundi	rn-Latn-BI	Burundi	2,261,931	1,130,966	regional_depth	MV-1 (D2)	AFG-S011
65	Maithili Nepal profile	mai-Deva-NP	Nepal	3,222,389	1,105,279	regional_depth	MV-1 (D2)	PM-S004
66	Indonesia Betawi Latin-script edition	bew-Latn-ID	Indonesia	2,244,648	1,077,431	regional_depth	MV-1 (D3)	PM-S005
67	South Africa siSwati Latin-script edition	ss-Latn-ZA	South Africa	1,692,719	771,457	regional_depth	MV-1 (D3)	ZA-S001
68	Ethiopia Wolaytta Latin-script edition	wal-Latn-ET	Ethiopia	1,627,955	492,131	regional_depth	MV-1 (D2)	PM-S020
69	Zimbabwean Ndebele	nd-Latn-ZW	Zimbabwe	1,599,324	745,525	regional_depth	MV-1 (D2)	PM-S011
70	South Africa Tshivenda Latin-script edition	ve-Latn-ZA	South Africa	1,480,565	674,768	regional_depth	MV-1 (D3)	ZA-S001
71	Bhojpuri Nepal profile	bho-Deva-NP	Nepal	1,820,795	627,264	regional_depth	MV-1 (D2)	PM-S004
72	Ethiopia Hadiyya Latin-script edition	hdy-Latn-ET	Ethiopia	1,253,894	379,052	regional_depth	MV-1 (D2)	PM-S020
73	Mauritius Mauritian Creole Latin-script edition	mfe-Latn-MU	Mauritius	1,045,558	492,917	regional_depth	MV-1 (D3)	PM-S013
74	South Africa isiNdebele Latin-script edition	nr-Latn-ZA	South Africa	1,044,377	475,975	regional_depth	MV-1 (D3)	ZA-S001
75	Indonesia Nias Latin-script edition	nia-Latn-ID	Indonesia	747,168	358,641	regional_depth	MV-1 (D3)	PM-S005

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
76	Standard Haitian Creole	ht-Latn-HT	United States	867,500	216,875	regional_depth	MV-1 (D3)	GRG-S001
77	Avadhi Nepal profile	awa-Deva-NP	Nepal	864,276	276,568	regional_depth	MV-1 (D2)	PM-S004
78	Silesian in ślabikŏrzowy szrajbōnek	szl-Latn-PL	Poland	467,145	116,786	regional_depth	MV-1 (D3)	GRG-S006
79	Zimbabwe Ndau Latin-script edition	ndc-Latn-ZW	Zimbabwe	372,607	173,691	regional_depth	MV-1 (D2)	PM-S011
80	Senegal Mandinka (Màndienka), official Latin profile	mnk-Latn-SN	Senegal	465,052	117,100	regional_depth	MV-1 (D2)	AFG-S009
81	Tetum Prasa	tet-Latn-TL	Timor-Leste	361,027	125,276	regional_depth	MV-1 (D2)	ALG-S014
82	YIVO Standard Yiddish edition for the explicitly named YIVO-reading audience	ydd-Hebr-US	United States	194,800	48,700	regional_depth	MV-1 (D3)	GRG-S001
83	English-oriented Pennsylvania German spelling used by Vitt Du Deitsh Shvetza / Di Heilich Shrift / Ich Kann PA Deitsh Shreiva	pdc-Latn-US	United States	185,200	46,300	regional_depth	MV-1 (D3)	GRG-S001
84	Diné Bizaad in Navajo Nation school orthography	nv-Latn-US	United States	161,200	40,300	regional_depth	MV-1 (D3)	GRG-S001
85	Timor-Leste Makasai Latin-script edition	mkz-Latn-TL	Timor-Leste	123,840	42,972	regional_depth	MV-1 (D2)	ALG-S014
86	New Zealand Samoan Latin-script edition	sm-Latn-NZ	New Zealand	110,541	27,635	regional_depth	MV-1 (D3)	PM-S014
87	Peru Ashaninka Latin-script edition	cni-Latn-PE	Peru	73,567	34,451	small_population_prestige_domain_screen	MV-1 (D2)	PM-S007
88	School-standard Kashubian	csb-Latn-PL	Poland	89,198	22,300	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S006
89	Peru Awajun Latin-script edition	agr-Latn-PE	Peru	56,584	26,498	small_population_prestige_domain_screen	MV-1 (D2)	PM-S007
90	Jamaican Creole in Cassidy/JLU orthography	jam-Latn-JM	United States	72,000	18,000	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S001
91	Timor-Leste Kemak Latin-script edition	kem-Latn-TL	Timor-Leste	68,995	23,941	small_population_prestige_domain_screen	MV-1 (D2)	ALG-S014

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Next allocation	Source ID
92	Timor-Leste Bunak Latin-script edition	bfn-Latn-TL	Timor-Leste	64,686	22,446	small_population_prestige_domain_screen	MV-1 (D2)	ALG-S014
93	Timor-Leste Tokodede Latin-script edition	tkd-Latn-TL	Timor-Leste	46,784	16,234	small_population_prestige_domain_screen	MV-1 (D2)	ALG-S014
94	Fataluku	ddg-Latn-TL	Timor-Leste	41,500	14,401	small_population_prestige_domain_screen	MV-1 (D2)	ALG-S014
95	Peru Shipibo-Konibo Latin-script edition	shp-Latn-PE	Peru	34,152	15,993	small_population_prestige_domain_screen	MV-1 (D2)	PM-S007
96	Standard ʻŌlelo Hawaiʻi with ʻokina and kahakō	haw-Latn-US	United States	32,730	8,182	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S001
97	Timor-Leste Waima'a Latin-script edition	wmh-Latn-TL	Timor-Leste	21,227	7,366	small_population_prestige_domain_screen	MV-1 (D2)	ALG-S014
98	Standard Marshallese under the 2010 Orthography Act	mh-Latn-MH	United States	27,160	6,790	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S001
99	Standard Central Alaskan Yup'ik	esu-Latn-US	United States	16,120	4,030	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S001
100	Chuuk State standard Chuukese	chk-Latn-FM	United States	14,700	3,675	small_population_prestige_domain_screen	MV-1 (D3)	GRG-S001

Appendix C. Ordered accessibility safeguard portfolio

Only safeguard 1 has foundational priority. Items 2–11 retain the deterministic registered backlog order and are not evidence-based cardinal ranks. Every access-gain interval has a conservative zero floor or no defensible denominator. Token increments are FR-2 reference sensitivities, not observed usage, billing, or whole-programme cost.

Order	Safeguard	Barrier	Stratum	Source ceiling	Gain status	FR-2 base-token increment	Product	Nonadditivity	Population source
1	Semantic HTML with accessible math	visual_disability	India	771288	zero_to_nested_ceiling_only	610957	Born-accessible localized HTML; semantic headings tables language metadata MathML intent/structure skip links keyboard navigation and tested screen-reader reading order	ACCPOP-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
2	Formula narration and synchronized audio	visual_disability	India	771288	zero_to_nested_ceiling_only	1425567	Deterministic language-specific symbol and formula narration synchronized with navigable text; diagrams require separate descriptions	ACCPOP-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
3	English mathematical Braille derivative	braille_math	United States and Canada		unbounded_unknown		Source-linked BRF using the jurisdictionally correct UEB technical or UEB-with-Nemeth policy; tactile/diagram alternatives; round-trip formula QA	visual impairment and screen-reader populations	ACC-S012
4	Localized British Sign Language instructional video and interactive glossary	signed_language	Scotland	117349	zero_to_language_ability_ceiling_only		Native BSL presentation with captions transcript downloadable low-bandwidth video glossary diagrams and source-aligned segment identifiers	none	ACC-S006
5	Logic figure and proof-tree remediation	visual_disability	India	771288	zero_to_nested_ceiling_only	346209	Programmatic proof structure with explicit subproof starts ends indentation rule names symbol pronunciation and linearized fallback	ACCPOP-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
6	Mathematics figure remediation	visual_disability	India	771288	zero_to_nested_ceiling_only	610957	Structured data plus concise alt text and adjacent long descriptions; non-colour-only encodings; downloadable data where meaningful	ACCPOP-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004

Order	Safeguard	Barrier	Stratum	Source ceiling	Gain status	FR-2 base-token increment	Product	Nonadditivity	Population source
7	Localized New Zealand Sign Language instructional video and interactive glossary	signed_language	New Zealand	24680	zero_to_language_ability_ceiling_only		Native NZSL presentation with captions transcript downloadable low-bandwidth video glossary diagrams and source-aligned segment identifiers	none	ACC-S005
8	Offline HTML EPUB or SCORM package	connectivity	Global	2200000000	zero_to_global_ceiling_only	346209	Dependency-free static HTML plus conformant EPUB; local assets search navigation checksums and print profile; no mandatory account analytics CDN font or runtime requests	every language territory and learner cell; all other delivery-axis interventions	ACC-S002
9	Tagged searchable screen and print PDF	visual_disability	India	771288	zero_to_nested_ceiling_only	305479	Tagged reading order bookmarks language metadata embedded fonts searchable text black-and-white print profile and formula alternatives	ACCPOP-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
10	Plain-language companion	cognitive_access_proxy	India	257755	zero_to_nested_proxy_ceiling_only	1934698	Aligned summaries definitions one-idea steps worked examples explicit goals consistent vocabulary and source links; preserve mathematical truth conditions	ACCPOP-COG-INDIA-5-19-2011; AX-PLAIN edges	ACC-S004
11	Localized Peruvian Sign Language instructional video and interactive glossary	signed_language	Peru	10447	zero_to_childhood_language_ceiling_only		Native Peruvian Sign Language presentation with Spanish captions transcript downloadable low-bandwidth video glossary diagrams and source-aligned segment identifiers	none	ACC-S007

Appendix D. Existing-work reconciliation and deficit effect

D1. Exact current local baselines

The table below is the bounded successor-state census used to avoid commissioning a duplicate full Open Logic edition. It is separate from the 209-row candidate register: a zero in that register's `covered` state means covered baselines were not proposed again, not that no complete local editions exist.

Exact profile or architecture	Current bounded state	Marginal treatment	Controlling asset / SHA-256
zh-Hans-CN	Complete Open Logic; OpenStax Algebra and Trigonometry 2e 94/94; Calculus Volume 1 3/55 at the frozen state	No duplicate Open Logic commission; remaining OpenStax scope only	ACE-A006 / 89B03BB2854AE28DCBA29B73887FA20B46296DC94B8077C21CF90866439D331D
hi-Deva-IN	Complete, published Open Logic with public-byte readback	Baseline for Hindi; never credited to Urdu or other South Asian profiles	ACE-A007 / C155255517CA9B57B9556ABD18CADB8A4E31E52725934F8A090C25D8C7D3B1C5
tr-TR	Complete Open Logic	Baseline for Turkish only; no automatic Turkic-family coverage	ACE-A008 / 8119903A18489AFEBEB18A1A787020869BB4F11E32C008C8849D351392B77834
ar (Modern Standard Arabic formal edition)	Complete, published Open Logic with public-byte readback	Baseline formal Arabic coverage; dialect, Amazigh, Kurdish, Nubian, South Arabian, signed, and accessibility needs remain separate	ACE-A009 / 8F0DB97861BE0777A49359E865531AC2E6DBF42B1BFA4D550A8817A000CC914B
fa-Arab-IR	Complete, published Iranian Persian Open Logic with public-byte readback	Baseline for Iranian Persian only; Dari, Tajik, Hazaragi, and other Iranic profiles remain separate	ACE-A009 / 8F0DB97861BE0777A49359E865531AC2E6DBF42B1BFA4D550A8817A000CC914B
es	Complete local Open Logic QA state	Baseline for exact Spanish output; no pan-Romance reach	ACE-A010 / F2A940018EB87B959112B71EE8435D23C24469925728CBFDDB8AF15717B4AFD0
pt-BR	Complete local Open Logic QA state	Baseline for Brazilian Portuguese only; no European/African Portuguese or pan-Romance reach	ACE-A010 / F2A940018EB87B959112B71EE8435D23C24469925728CBFDDB8AF15717B4AFD0

Exact profile or architecture	Current bounded state	Marginal treatment	Controlling asset / SHA-256
id-ID	Partial Open Logic: 321/722 units and 181,791/367,220 source alpha tokens	401 units and 185,429 tokens remain; token-weighted deficit D = 0.5049501669	ACE-A011 / 990D23F7168A5AB09F6FCB595B5F9BFA1114B36D63DF6EE0282A5C72C7720211; ACE-A092 / 26EB3FDB4098182C1AD4F3245A673556B4AFAABFD3AAE5B40DA88F5DDAEA418B
isv-Latn / isv-Cyrl programme	Seven accepted Open Logic units of 722; no finished reader	Unit progress only; no scalar token deficit or cross-language demographic reach inferred	ACE-A012 / 8515634D615773223074590F0DB4F2612BDF564BDDEC8E8015A6D6BB7815D3FB
Proposed Romance constructed surface	722 provisional files but zero canon-admitted units	Research/architecture hypothesis only; zero cardinal reach	ACE-A010 / F2A940018EB87B959112B71EE8435D23C24469925728CBFDDB8AF15717B4AFD0

D2. Candidate-register reconciliation states

State	Rows	Registered meaning	Deficit effect	Boundary
covered	0	Exact declared target scope is registered as delivered and verifiable.	D=0 only for the exact verified target × unit × format; any uncovered curriculum or format remains D=1.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.
partial	18	At least one exact unit/component is registered, but the declared target scope is incomplete or inconsistent.	D=0 only on verified covered units/components and D=1 on the residual; no scalar is assigned without a common exact denominator.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.
researched	2	Research exists, but no delivered target coverage is registered.	Research without delivered target bytes does not reduce the content deficit; D=1 for the missing comparator.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.
dormant	0	A prior item is registered as dormant and lacks current usable/build evidence.	A dormant item is not credited without verified usable current bytes and build identity; D=1 pending that registered evidence.	No rows in the current 209-row register meet this state; retained as an explicit fail-closed rule.
duplicated	0	The same usable target coverage is registered more than once; only a canonical copy may count.	The canonical item counts once; a duplicate adds no marginal coverage and produces no further reduction in D.	No rows in the current 209-row register meet this state; retained as an explicit fail-closed rule.

State	Rows	Registered meaning	Deficit effect	Boundary
missing	189	No exact target coverage is registered in the bounded local census.	No registered target coverage: D=1 for the declared target × unit × format.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.

Indonesian is the only partial target with a common exact token denominator: 321 of 722 units contain 181,791 of 367,220 source alpha tokens, leaving 185,429 tokens and a token-weighted deficit of 0.5049501669. Interslavic has seven accepted units, but no comparable accepted-source-token denominator is registered, so this paper does not assign it a scalar deficit. All other partial rows remain component-level. The detailed 209-row reconciliation is machine-readable in `appendix_a_existing_work_reconciliation.csv`.

Appendix E. Global regional, source, measure, and confidence gap map

Region	Subregion	Cardinal rows	Top100	Sources	Measures	High	Medium	Aggregation rule
Africa	Eastern Africa	1	1	1	derived_census_any_kirundi_read_and_written_persons_age_10_plus	1	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Horn of Africa	6	6	1	census_mother_tongue_persons	6	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Indian Ocean	1	1	1	language_usually_spoken_at_home_persons	1	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Southern Africa	12	12	2	census_language_most_often_spoken_in_household_persons_age_1_plus || census_mother_tongue_persons	12	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Western Africa	4	4	2	census_mother_tongue_persons_age_3_plus || census_principal_language_commonly_spoken_persons_age_3_plus	3	1	not summed: measures, years, territories, and overlap universes may differ
Americas	Arctic	1	0	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	1	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America	13	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	13	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Caribbean and Central American diaspora)	1	0	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	1	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Caribbean diaspora)	3	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	3	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Central American Indigenous diaspora)	4	0	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	4	0	not summed: measures, years, territories, and overlap universes may differ

Region	Subregion	Cardinal rows	Top100	Sources	Measures	High	Medium	Aggregation rule
Americas	North America (European nonterritorial diaspora)	2	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Pacific diaspora)	8	3	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	8	0	not summed: measures, years, territories, and overlap universes may differ
Americas	South America	3	3	1	census_language_learned_in_childhood_age3plus_persons	3	0	not summed: measures, years, territories, and overlap universes may differ
Asia	Central Asia	5	5	5	census_mother_tongue_persons || census_native_language_of_own_nationality_persons || census_native_language_persons || derived_cldr_territory_functional_language_users || preliminary_census_mother_tongue_persons_rounded	4	1	not summed: measures, years, territories, and overlap universes may differ
Asia	East Asia	1	1	1	derived_cldr_territory_functional_language_users	0	1	not summed: measures, years, territories, and overlap universes may differ
Asia	South Asia	26	26	5	census_mother_tongue_persons || census_same_name_mother_tongue_component_persons || derived_2020_total_speakers_from_official_secondary_share_and_world_bank_denominator || derived_cldr_territory_functional_language_users	21	5	not summed: measures, years, territories, and overlap universes may differ
Asia	Southeast Asia	25	25	7	census_mother_tongue_persons || CLEAR_normalized_main_language_weighted_persons_from_IPUMS_2010_census_extract || daily_language_at_home_age5plus_persons || derived_cldr_territory_functional_language_users || derived_union_of_mutually_exclusive_census_strata_persons || peer_reviewed_published_lower_bound_speakers	22	3	not summed: measures, years, territories, and overlap universes may differ
Europe	Central Europe	4	4	1	census_languages_used_at_home_multiple_response_persons	4	0	not summed: measures, years, territories, and overlap universes may differ
Europe	Northern Europe	1	1	1	self_reported_daily_use_outside_education_persons	1	0	not summed: measures, years, territories, and overlap universes may differ

Region	Subregion	Cardinal rows	Top100	Sources	Measures	High	Medium	Aggregation rule
Oceania	Australia	10	0	4	census_language_used_at_home_persons	10	0	not summed: measures, years, territories, and overlap universes may differ
Oceania	Polynesia	2	2	1	census_languages_spoken_multiple_response_perso ns	2	0	not summed: measures, years, territories, and overlap universes may differ

The global detail table has one row for each of the 133 cardinal natural-language interventions and an exact join to one population observation and its registered source. The source/measure/confidence table preserves source-level groupings. Counts in this appendix describe records and coverage strata; they are not summed population claims.

Appendix F. Curriculum portfolios, adaptation depths, and Top-100 mapping

F1. Exact curriculum portfolios

ID	Portfolio	Project	Exact content	Units	Source tokens	Preferred depth	Prerequisite	Sources	Caveat
MV-1	Minimum viable quantitative curriculum	OpenStax	Algebra and Trigonometry 2e	94		D2		ACE-E002;ACE-E003	Whole book retained because later statistics physics and calculus reuse its sequence and vocabulary.
SB-1	Secondary data bridge	OpenStax	MV-1 plus Introductory Statistics 2e	196		D2	MV-1	ACE-E002;ACE-E003	Statistics chapters are not fully independent; partial extracts must be labeled as packets.
SB-2	Secondary STEM bridge	OpenStax	SB-1 plus College Physics 2e	479		D2_then_D3	SB-1	ACE-E002;ACE-E003	College Physics is algebra based and does not require calculus.
UG-1	Undergraduate STEM breadth	OpenStax	SB-2 plus Chemistry 2e and Biology 2e	887		D2	SB-2	ACE-E002;ACE-E003	Terminology figure and alt-text burden rises substantially.
UG-2	Complete undergraduate STEM core	OpenStax	UG-1 plus Calculus Volumes 1 through 3	1050		D2_then_D3	UG-1	ACE-E002;ACE-E003	Calculus volumes must be sequenced 1 then 2 then 3.
FR-1	Formal-reasoning foundation	Open Logic Project	methods plus sets-functions-relations plus propositional	87	50330	D2		ACE-A013;ACE-E004;ACE-E005	Compact mathematical-reasoning route.
FR-2	Formal-reasoning core	Open Logic Project	FR-1 plus first-order-logic	210	120083	D3	FR-1	ACE-A013;ACE-E004;ACE-E005	Common complete-edition comparator used by the cardinal portfolio.
FR-3	Logic and computation bridge	Open Logic Project	FR-2 plus computability plus turing-machines	276	157048	D2	FR-2	ACE-A013;ACE-E004	Suitable as a logic/computer-science bridge.
FR-4	Advanced formal-reasoning portfolio	Open Logic Project	full canonical reader	642	318916	D1_or_D2	FR-2	ACE-A013;ACE-E004	Select by official remix rather than repository order.
FR-5	Complete archival corpus	Open Logic Project	canonical plus supplements	722	367220	D1_unless_selected	FR-4	ACE-A013;ACE-E004	Not automatically a teachable sequence.

ID	Portfolio	Project	Exact content	Units	Source tokens	Preferred depth	Prerequisite	Sources	Caveat
AX-1	Core accessibility package	Cross-project	semantic HTML or MathML plus screen PDF print PDF and offline package			parallel_to_D2		ACE-E003;ACE-E005	Apply to every selected language and curriculum tier rather than deferring.
AX-2	Enhanced nonvisual access	Cross-project	formula narration diagram descriptions and audio or TTS-ready text			separate_derivative	AX-1	ACE-E003;ACE-E005	Requires deterministic mathematical and chemistry reading conventions.
AX-3	Plain-language companion	Cross-project	definitions summaries and worked-example explanations			separate_derivative	AX-1	ACE-E003;ACE-E005	Supplement only; never silently replaces the source-faithful edition.

F2. Exact adaptation depths

ID	Depth	Included components	Low	Base	High	Status	Caveat
D0	Terminology and structure	TOC metadata learning objectives key-term graph notation registry attribution	0.05	0.075	0.10	infrastructure	Not an educational edition.
D1	Reading edition	Expository narrative definitions summaries captions and a small worked-example set	0.30	0.375	0.45	reference_or_early_reach	Bulk assessment and practice architecture omitted.
D2	Course-ready core	Complete exposition definitions worked examples checkpoints representative exercises and answers figures and alt text	0.60	0.70	0.80	minimum_course_intervention	Default first educational production depth.
D3	Complete public edition	Every public module component exercise public answer caption glossary reference and attribution	1.00	1.00	1.00	complete_public_source	Does not include restricted instructor-only ancillaries.
D4	Local pedagogical adaptation	D3 plus locally appropriate data units software instructions examples and explanatory scaffolding	1.20	1.40	1.60	localized_complete	Separate from source-faithful translation and requires explicit adaptation labeling.

F3. Full Top100 target-by-curriculum mapping

This mapping intentionally omits the full score fields already present in `TOP_100.csv`. It preserves the ordered target identity and adds only the first-product and next-curriculum/depth assignments. The distribution is MV-1/D2 = 36, MV-1/D3 = 44, and SB-1/D3 = 20. SB-1 means **MV-1 plus Introductory Statistics 2e**.

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
1	NAT-002	Indian Bengali	bn-Beng-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
2	NAT-001	Bangladesh Bangla	bn-Beng-BD	high_reach_underserved	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
3	NAT-003	Telugu	te-Telu-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
4	NAT-006	Marathi	mr-Deva-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
5	NAT-028	Vietnamese	vi-Latn-VN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	ASSEC-S007
6	NAT-038	Javanese	jv-Latn-ID	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S005
7	NAT-004	Indian Tamil	ta-Taml-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
8	NAT-015	Western Punjabi	pnb-Arab-PK	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
9	NAT-007	Gujarati	gu-Gujr-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
10	NAT-008	Kannada	kn-Knda-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
11	NAT-029	Thai	th-Thai-TH	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	TH-S001
12	NAT-010	Odia	or-Orya-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
13	NAT-013	Indian Urdu	ur-Arab-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
14	NAT-030	Burmese	my-Mymr-MM	high_reach_underserved	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
15	NAT-014	Eastern Punjabi	pa-Guru-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
16	NAT-009	Malayalam	ml-Mlym-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
17	EXP-003	Uzbekistan Uzbek Latin-script edition	uz-Latn-UZ	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	ALG-S018
18	NAT-039	Sundanese	su-Latn-ID	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S005
19	NAT-034	Cebuano	ceb-Latn-PH	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	ASSEC-S008
20	NAT-018	Sindhi	sd-Arab-PK	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
21	NAT-065	Amharic	am-Ethi-ET	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S020

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
22	NAT-011	Assamese	as-Beng-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
23	NAT-058	Dari	prs-Arab-AF	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ASSEC-S018
24	NAT-012	Pakistani Urdu	ur-Arab-PK	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
25	NAT-019	Saraiki	skr-Arab-PK	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
26	EXP-001	Afghanistan Pashto Arabic-script edition	ps-Arab-AF	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ASSEC-S018
27	NAT-027	Sinhala	si-Sinh-LK	high_reach_underserved	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
28	EXP-004	India Maithili Devanagari edition	mai-Deva-IN	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
29	NAT-031	Khmer	km-Khmr-KH	high_reach_underserved	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ALG-S011
30	NAT-022	Nepali	ne-Deva-NP	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
31	NAT-073	isiZulu	zu-Latn-ZA	high_reach_underserved	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
32	NAT-047	Kazakh	kk-Cyrl-KZ	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	ALG-S007
33	NAT-055	Turkmen	tk-Latn-TM	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ALG-S009
34	NAT-075	Shona	sn-Latn-ZW	high_reach_underserved	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S011
35	NAT-049	Tajik	tg-Cyrl-TJ	high_reach_underserved	FR-2	D3	SB-1	D3	MV-1 plus Introductory Statistics 2e	ASSEC-S006

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
36	NAT-032	Lao	lo-Laoo-LA	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ASSEC-S006
37	NAT-074	isiXhosa	xh-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
38	EXP-018	Mali Bamanankan Latin-script edition	bm-Latn-ML	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	AFG-S008
39	EXP-013	Ethiopia Somali Latin-script edition	so-Latn-ET	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
40	EXP-005	Indonesia Madurese Latin-script edition	mad-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
41	NAT-081	Wolof	wo-Latn-SN	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	AFG-S009
42	NAT-048	Kyrgyz	ky-Cyrl-KG	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ALG-S008
43	NAT-035	Ilocano	ilo-Latn-PH	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S008
44	NAT-036	Hiligaynon	hil-Latn-PH	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S008
45	NAT-005	Sri Lankan Tamil	ta-Taml-LK	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
46	EXP-019	South Africa Sepedi Latin-script edition	nso-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
47	EXP-020	South Africa Setswana Latin-script edition	tn-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
48	NAT-050	Mongolian	mn-Cyrl-MN	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
49	EXP-021	South Africa Sesotho Latin-script edition	st-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
50	EXP-014	Ethiopia Tigrinya Ethiopic-script edition	ti-Ethi-ET	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S020

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
51	NAT-105	Maori	mi-Latn-NZ	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S014
52	EXP-006	Indonesia Minangkabau Latin-script edition	min-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
53	OBS-POP-SN-AFG-002	Senegal Pulaar, official Latin orthography	fuc-Latn-SN	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	AFG-S009
54	EXP-037	Ireland Irish Latin-script daily-use edition	ga-Latn-IE	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S016
55	EXP-007	Indonesia Banjar Latin-script edition	bjn-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
56	EXP-015	Ethiopia Sidama Latin-script edition	sid-Latn-ET	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
57	OBS-GRG-PL-007	Official-standard Belarusian	be-Cyrl-BY	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
58	NAT-037	Waray	war-Latn-PH	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S008
59	EXP-022	South Africa Xitsonga Latin-script edition	ts-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
60	OBS-GRG-PL-010	Standard Lithuanian	lt-Latn-LT	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
61	EXP-008	Indonesia Sasak Latin-script edition	sas-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
62	NAT-021	Brahui	brh-Arab-PK	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
63	EXP-009	Indonesia Acehnese Latin-script edition	ace-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
64	NAT-079	Kirundi	rn-Latn-BI	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	AFG-S011
65	NAT-023	Maithili Nepal profile	mai-Deva-NP	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
66	EXP-010	Indonesia Betawi Latin-script edition	bew-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
67	EXP-023	South Africa siSwati Latin-script edition	ss-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
68	EXP-016	Ethiopia Wolaytta Latin-script edition	wal-Latn-ET	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
69	NAT-076	Zimbabwean Ndebele	nd-Latn-ZW	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S011
70	EXP-024	South Africa Tshivenda Latin-script edition	ve-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
71	NAT-024	Bhojpuri Nepal profile	bho-Deva-NP	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
72	EXP-017	Ethiopia Hadiyya Latin-script edition	hdy-Latn-ET	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
73	EXP-029	Mauritius Mauritian Creole Latin-script edition	mfe-Latn-MU	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S013
74	EXP-025	South Africa isiNdebele Latin-script edition	nr-Latn-ZA	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
75	EXP-011	Indonesia Nias Latin-script edition	nia-Latn-ID	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
76	OBS-GRG-US-025	Standard Haitian Creole	ht-Latn-HT	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
77	NAT-026	Avadhi Nepal profile	awa-Deva-NP	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
78	OBS-GRG-PL-002	Silesian in ślabikŏrzowy szrajbōnek	szl-Latn-PL	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
79	EXP-030	Zimbabwe Ndau Latin-script edition	ndc-Latn-ZW	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S011
80	OBS-POP-SN-AFG-005	Senegal Mandinka (Màndienka), official Latin profile	mnk-Latn-SN	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	AFG-S009

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
81	NAT-041	Tetum Prasa	tet-Latn-TL	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
82	OBS-GRG-US-030	YIVO Standard Yiddish edition for the explicitly named YIVO-reading audience	ydd-Hebr-US	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
83	OBS-GRG-US-029	English-oriented Pennsylvania German spelling used by Vitt Du Deitsh Shvetza / Di Heilich Shrift / Ich Kann PA Deitsh Shreiva	pdc-Latn-US	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
84	OBS-GRG-US-003	Diné Bizaad in Navajo Nation school orthography	nv-Latn-US	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
85	EXP-031	Timor-Leste Makasai Latin-script edition	mkz-Latn-TL	regional_depth	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
86	EXP-036	New Zealand Samoan Latin-script edition	sm-Latn-NZ	regional_depth	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	PM-S014
87	EXP-026	Peru Ashaninka Latin-script edition	cni-Latn-PE	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S007
88	OBS-GRG-PL-004	School-standard Kashubian	csb-Latn-PL	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
89	EXP-027	Peru Awajun Latin-script edition	agr-Latn-PE	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S007
90	OBS-GRG-US-026	Jamaican Creole in Cassidy/JLU orthography	jam-Latn-JM	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
91	EXP-032	Timor-Leste Kemak Latin-script edition	kem-Latn-TL	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
92	EXP-033	Timor-Leste Bunak Latin-script edition	bfn-Latn-TL	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
93	EXP-034	Timor-Leste Tokodede Latin-script edition	tkd-Latn-TL	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
94	NAT-044	Fataluku	ddg-Latn-TL	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
95	EXP-028	Peru Shipibo-Konibo Latin-script edition	shp-Latn-PE	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	PM-S007

Pos.	ID	Target	Profile	Lane	First	First depth	Next	Next depth	Next exact content	Population source
96	OBS-GRG-US-023	Standard ʻŌlelo Hawaiʻi with ʻokina and kahakō	haw-Latn-US	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
97	EXP-035	Timor-Leste Waima'a Latin-script edition	wmh-Latn-TL	small_population_prestige_domain_screen	FR-2	D3	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
98	OBS-GRG-US-017	Standard Marshallese under the 2010 Orthography Act	mh-Latn-MH	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
99	OBS-GRG-US-001	Standard Central Alaskan Yup'ik	esu-Latn-US	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
100	OBS-GRG-US-018	Chuuk State standard Chuukese	chk-Latn-FM	small_population_prestige_domain_screen	FR-2	D3	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001

Appendix G. Unresolved output profiles and D0 exclusions

ID	Issue	Target	Profile	Source population	Measure	Year	Population source	Why excluded	Evidence sought
EXP-002	D0_profile_population_mismatch	Pakistani Pashto	ps-Arab-PK	43633946	census_mother_tongue_persons	2023	PM-S003	exclude_D0_profile_population_mismatch; Pakistan census label Pushto is broader than an exact production variety	Exact production-variety/standard mapping within the PBS Pushto umbrella and a population-to-profile join that does not assign the whole category to one variety.
EXP-012	D0_profile_population_mismatch	Oromo	om-Latn-ET	24930424	census_mother_tongue_persons	2007	PM-S020	exclude_D0_profile_population_mismatch; census Oromigna category is broad and its editorial identifier is orm:CSA; exact variety reach is unresolved	Exact production-variety/standard mapping within the CSA Oromigna category, with cross-border populations excluded unless separately and disjointly evidenced.
OBS-GRG-US-004	unresolved_localized_output_profile	Eastern Keres	kee-Zzzz-US target profile; script unresolved by population source	12540	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2017-2021	GRG-S001	not rankable as one edition; split into community profiles first	Community-specific Eastern Keres variety, script, orthography, educational-standard authority, and a nonmultiplying population-to-output join.
OBS-GRG-US-012	unresolved_localized_output_profile	Tewa	tew-Zzzz-US target profile; script unresolved by population source	5105	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2017-2021	GRG-S001	not rankable as one edition; split into community profiles first	Pueblo-specific Tewa variety, script, orthography, educational-standard authority, and a nonmultiplying population-to-output join.
OBS-GRG-US-027	unresolved_localized_output_profile	Guyanese Creole English	gyn-Latn-GY	4603	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2017-2021	GRG-S001	rank shared core and localized profiles separately; do not multiply one population observation	Exact institutional teaching orthography and an audience split between a shared core and localized profiles without multiplying GRG-US-027.

These are exclusions from recommendation ranking, not claims that the communities lack educational need. The two D0 rows retain source population cells but do not assign them to an exact production profile.

Appendix H. Interlanguage overlap matrix summary

The complete registered matrix contains 104 rows: 80 interlanguage rows across 15 intervention IDs and 24 accessibility rows across 18 IDs. IL-AR occurs in both mechanism scopes, so those ID counts are nonadditive and reconcile to 32 distinct IDs overall. Appendix F summarizes only the 80 interlanguage rows; Appendix C handles accessibility. Every interlanguage component remains non-rankable under current evidence, and every demographic or component subtotal is nonadditive unless an explicit disjoint-universe rule says otherwise.

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-AR	pluricentric_shared_core	1	Readers comfortable with Modern Standard Arabic	Arab	none registered for target corpus	formal literacy required	complete local Arabic edition	Amazigh Kurdish Nubian Modern South Arabian and signed languages	Existing MSA coverage is baseline and cannot be claimed again by scaffolds
IL-BCMS	dual_script_register || pluricentric_shared_core	5	Bosnian-standard cell || Croatian-standard cell || Montenegrin-standard cell || Serbian Cyrillic-reading stratum || Serbian Latin-reading stratum	Cyrl || Latn	none registered	none beyond named-standard literacy || none beyond script literacy	none established	Croatian Montenegrin Serbian and non-BCMS languages remain distinct cells || Other BCMS standards || Other BCMS standards remain distinct cells	Named output claims only Bosnian cell; shared source has compute value only || Named output claims only Croatian cell || Named output claims only Montenegrin cell || Partition one Serbian population by script use; do not add Latin and Cyrillic

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-GERM	natural_intercomprehension_reuse	10	Afrikaans-standard cell || Dutch-standard cell || Faroese cell || Hutterisch cell || Low German exact-standard cell || Pennsylvania Dutch cell || Plautdietsch cell || Scots cell || West Frisian cell || Yiddish exact-register cell	Hebr || Latn	none	none beyond community-standard literacy || none beyond exact-register literacy || none beyond exact-standard literacy || none beyond named-standard literacy	Danish and English overlap unmeasured || Dutch and English overlap unmeasured || English and German overlap unmeasured || English Hebrew Russian and German overlaps vary || English overlap expected high but unmeasured || English overlap expected very high but unmeasured || English overlap likely high but unmeasured || German English Russian Spanish and Portuguese overlaps vary || German overlap unmeasured	All Latin-script Germanic profiles || All residual Germanic communities || Dutch Afrikaans and other Germanic communities || Mainland Scandinavian and other Germanic languages || Other Frisian languages and Germanic communities || Other Germanic communities || Other Germanic diaspora languages || Other Germanic languages	Claims only residual Afrikaans cell; directional Dutch credit zero || Claims only residual Dutch cell; cross-language credit zero || Count only residual comfortable-reader gain; no pan-Germanic credit || Count only residual community readers; no pan-Germanic credit || Deduplicate by residence and dominant academic-language access; no pan-Germanic credit || No pan-Germanic credit || No pan-Germanic credit; named output only || No pan-Germanic credit; residence and community register define separate cells || No pan-Germanic or mainland Scandinavian credit

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-HU	dual_script_register	3	Indian Urdu cell || Pakistani Urdu cell || Standard Hindi-oriented cell	Arab Nastaliq || Deva	none	none beyond named-standard literacy	C-17 reports English-or-Hindi subsidiary-language knowledge for 23172258 of 50772631 Urdu-category speakers; academic comfort remains unknown || complete local Hindi edition; C-17 reports English subsidiary-language knowledge for 35257144 of the 528347193 Hindi census umbrella || Hindi overlap unmeasured	Other Indian mother tongues || Punjabi Sindhi Saraiki Pashto and other languages || Urdu and Hindi census umbrella sublanguages	Existing Hindi coverage is baseline and not added to Urdu adaptation reach; C-17 language knowledge is not academic comfort || Hindi gives zero Urdu credit; named Urdu output claims exact Pakistan cell || Keep Indian Urdu separate from Pakistan Urdu; apply disclosed nonoverlap sensitivity 0.543607303/0.771803651/0.908721461 rather than categorical Hindi coverage
IL-IDMS	dual_script_register || pluricentric_shared_core	4	Brunei Malay educational-standard cell || Indonesian standard cell || Malaysian Malay Jawi-reading stratum || Malaysian Malay Rumi cell	Arab || Latn	none	none beyond exact-standard literacy || none beyond Jawi literacy || none beyond named-standard literacy	321 of 722 Open Logic units local || Indonesian and Malaysian overlap untested || Indonesian cross-standard overlap untested || Same language as ms-Latn-MY	Javanese Sundanese Madurese Minangkabau Acehnese and other languages || Other Malayic and Indonesian languages || Other Malayic languages	Current Indonesian progress is subtracted; no Malaysian reach || Named Brunei cell only; no regional blanket credit || Partition or model dual literacy; never add Jawi and Rumi populations || Zero Indonesian credit pending directional technical-text audit; localized output claims Malaysian cell

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-ISV	constructed_bridge	15	Belarusian-literate cohort || Bosnian-standard cohort || Bulgarian-literate cohort || Croatian-standard cohort || Czech-literate cohort || Macedonian-literate cohort || Montenegrin-standard cohort || Polish-literate cohort || Pooled native-Slavic study respondents || Russian-literate cohort || Serbian Cyrillic-script reading cell || Serbian Latin-script reading cell || Slovak-literate cohort || Slovene-literate cohort || Ukrainian-literate cohort	Cyrl || Latn || study surface reported as written Interslavic	none at exact leaf level in current register || seven-gap written professional-text cloze [seven-gap short written professional-text cloze] score=0.84	not required for the short cloze; sustained mathematics unknown || unknown for sustained mathematics	No exact local native-language-edition overlap was subtracted || unknown || unknown in current local coverage census	Kashubian is not included || Kashubian Sorbian Rusyn and all untested minority leaves || No additive credit with Serbian Cyrillic || No additive credit with Serbian Latin || No BCMS or other South-Slavic transfer credit || No Bulgarian or other South-Slavic transfer credit || No Czech or other West-Slavic transfer credit || No Macedonian or other South-Slavic transfer credit || No Slovak or other West-Slavic transfer credit || No unnamed BCMS population aggregation || No untested minority or exact language leaf receives this pooled value || Untested Slavic leaves	Credit zero until an exact surface/script/cohort test; deduct any comfortable Russian edition before reach || Credit zero until exact testing and deduct any comfortable Ukrainian edition || Partition one Serbian population by observed script use; then deduct named Serbian access || The pooled 0.84 is one study-level point estimate and may not be copied to leaf rows or multiplied by a Slavic demographic ceiling || Zero until exact testing and after any named BCMS edition || Zero until exact testing and overlap subtraction || Zero until exact testing; deduct Czech and English comfortable access || Zero until exact testing; deduct Polish and English comfortable access || Zero until exact testing; deduct Slovak and English comfortable access || Zero until exact testing; deduct Slovene and English comfortable access || Zero until exact testing; one person cannot be counted in both Latin and Cyrillic projection cells

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-MANDING	dual_script_register || pluricentric_shared_core	5	Bambara Latin-standard cell || Bambara N'Ko-literate stratum || Jula Latin-standard cell || Maninka exact-national-standard cell || Maninka N'Ko-literate stratum	Latn || Nkoo	none registered	none beyond exact-standard literacy || none beyond N'Ko literacy || none beyond named-standard literacy	none established || Same Bambara language cell as Latin output || Same Maninka language cell as Latin output	Other Manding and Mande languages || Other Manding languages	Named output and country cell only || Named output and territory cell only || Named output only; no continuum-wide population || Partition by observed script use; never add Latin and N'Ko users
IL-NGUNI	pluricentric_shared_core	4	isiXhosa-standard cell || isiZulu-standard cell || siSwati-standard cell || Southern Ndebele standard cell	Latn	none registered	none beyond named-standard literacy	none established	isiXhosa siSwati isiNdebele and non-Nguni languages || Other Nguni standards || Zimbabwean Ndebele and other Nguni standards	Country cells and named output only || Keep nr and nd distinct; named output only || Named output only; no combined Nguni population || Named output only; no cross-language credit
IL-PDT	pluricentric_shared_core	3	Dari-standard cell || Iranian Persian cell || Tajik-standard cell	Arab || Cyrl	none	none beyond named-standard literacy	complete local Iranian Persian edition || Iranian Persian directional overlap unmeasured	Dari Tajik Kurdish Pashto Balochi and Pamiri languages || Pamiri Uzbek Russian and other languages || Pashto Hazaragi and other languages	Existing fa-IR coverage is baseline and not added to adaptation reach || Zero fa-IR reach pending technical-text evidence; named Dari output claims only exact cells || Zero fa-IR reach; named Tajik output only; transliteration is not additive
IL-PUNJABI	dual_script_register	2	Eastern Punjabi Gurmukhi cell || Western Punjabi Shahmukhi cell	Arab Shahmukhi || Guru	none	none beyond named-standard literacy	C-17 reports English-or-Hindi subsidiary-language knowledge for 17571723 of 33124726 Punjabi-category speakers; academic comfort remains unknown || none established	Saraiki Hindko and unresolved Punjabi census sublabels || Western Punjabi Saraiki Hindko	Claims only India Gurmukhi cell; apply nonoverlap sensitivity 0.469528503/0.734764251/0.893905701 and never add a dual-script population || Claims only resolved Shahmukhi population; no additive dual-script population

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-ROM	constructed_bridge || pluricentric_shared_core	8	Brazilian Portuguese cell || Catalan-standard cell || French-standard cell || Galician-standard cell || Italian-standard cell || Proposed constructed pan-Romance surface || Romanian-standard cell || Spanish-standard cell	Latn	none || none required for own localized output	none beyond literacy in exact locale standard || none beyond literacy in exact standard || none beyond literacy in named standard || none beyond literacy in the named standard || unknown	complete local edition || not established || not established in current successor census || Spanish and Brazilian Portuguese complete locally	All named Romance languages absent exact testing || Aromanian and other Eastern Romance varieties || Catalan Galician and other languages are not Spanish cells || European and African Portuguese profiles not automatically included || Occitan and other Romance languages || Other Romance languages || Portuguese and Spanish are separate baseline editions || Sardinian Sicilian Neapolitan Friulian and Ladin	Claims only Catalan cell; Spanish overlap deducted by cohort || Claims only exact French cell; bilingual overlap deducted separately || Claims only exact Romanian cell || Claims only Italian cell || Claims only residual Galician cell after measured Spanish/Portuguese comfort || Constructed surface has zero coverage; design artifacts cannot claim demographic ceilings || pt-BR output is subtracted from residual bridge scenarios; no additive bridge credit || Spanish output claims only exact Spanish residual cells and is subtracted before any bridge scenario
IL-SCAND	natural_intercomprehension_reuse	4	Danish-standard cell || Norwegian Bokmal cell || Norwegian Nynorsk cell || Swedish-standard cell	Latn	exact pair values not registered	none beyond named-standard literacy	Bokmal English and Scandinavian overlap unmeasured || English and Scandinavian overlap unmeasured	Faroese Icelandic || Nynorsk Faroese Icelandic	Named output only; cross-language reach zero until exact directional value is imported || Named output only; no cross-language credit || Partition Norwegian readers and do not add dual-standard literacy
IL-SOTHO	pluricentric_shared_core	3	Northern Sotho standard cell || Southern Sotho standard cell || Tswana standard cell	Latn	none registered	none beyond named-standard literacy	none established	Other Sotho-Tswana standards || Tswana Southern Sotho Venda Tsonga and Nguni languages	Country cells remain separate; named output only || Named output only; no combined Sotho-Tswana population

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-SQUECHUA	pluricentric_shared_core	3	Ayacucho Quechua cell || Cusco Quechua cell || South Bolivian Quechua cell	Latn	none registered	none beyond named-variety literacy	Spanish overlap unmeasured	Ayacucho South Bolivian Northern and Central Quechua || Cusco South Bolivian Northern and Central Quechua || Peruvian Southern Northern and Central Quechua	Named variety and country cell only; no aggregate Quechua population || Named variety only; no aggregate Quechua population
IL-TURKIC	natural_intercomprehension_reuse	10	Azerbaijani-standard cell || Bashkir-standard cell || Gagauz-standard cell || Kazakh-standard cell || Kyrgyz-standard cell || Tatar-standard cell || Turkish-standard cell || Turkmen-standard cell || Uyghur-standard cell || Uzbek-standard cell	Arab || Cyrl || Latn	none	none beyond named-standard literacy	complete local Turkish edition || Russian and Turkic overlap unmeasured || Turkish overlap unmeasured || Uzbek and Turkic overlap unmeasured	All other Turkic languages || Oghuz and other Kipchak languages || Other Karluk profiles || Other Kipchak languages || Other Oghuz languages || Uyghur and all other Karluk profiles	Cross-language reach zero; named output only || Cross-language Turkish credit is zero; named Azerbaijani output claims only Azerbaijani cell || No cross-language credit || No cross-language credit; legacy Cyrillic is a script-access cell || No cross-language credit; script projection is not additive || Turkish population is already covered locally and is never added to branch reuse reach

The Interslavic short-cloze evidence remains a task-specific observed result; it is not a sustained mathematics-comprehension estimate and is never multiplied by a Slavic-family demographic total.

Appendix I. Table-source authorities and source notes

Generated from the exact source IDs used by the current Table 2 and Top 100 snapshots. Draft references use only locally registered metadata; each entry retains its authorship basis and bounded source note.

Population and count authorities

AFG-S008

Institut National de la Statistique (INSTAT) and Bureau Central du Recensement (BCR), Mali. (2022). *RGPH5 2022: Caractéristiques culturelles de la population* [official census thematic report]. https://bibliostat.instat.ml/greenstone3/library/collection/dmograph/document/HASH01e98829c5a7e2847f521a98?ed=1

Authorship basis: registered authority used as corporate author

Source note: AFG-S008 supplies Resident population aged 3 years and above by mother-tongue category for Mali. Registered locator: Tableau 3.01, printed p.25; HTML document node HASH01e98829c5a7e2847f521a98.40. Registered temporal field: 2022. Caveat: Table 3.01 and Annex A03 publish materially different counts, and some printed sex subtotals do not exactly add to the printed ensemble cells. These observations preserve the Table 3.01 ensemble cells verbatim and do not repair or combine them. Peulh/Fulfulde and Malinké/Maninkakan remain source aggregates unless separately resolved.

AFG-S009

Agence Nationale de la Statistique et de la Démographie (ANSD), Senegal. (2023). *RGPH-5 2023 definitive report, Theme I: Etat et structure, urbanisation et caractéristiques socioculturelles de la population* [official census report]. https://www.ansd.sn/sites/default/files/recensements/rapport/rapport_national/Rapport-def-RGPH-5.pdf

Authorship basis: registered authority used as corporate author

Source note: AFG-S009 supplies Resident population aged 3 years and above by principal language commonly spoken for Senegal. Registered locator: Tableau I-32, printed pp.57-58; PDF pp.94-95. Registered temporal field: 2023. Caveat: Principal-language categories are mutually exclusive and do not measure all languages spoken, literacy, or academic reading ability. Joola is not a single exact variety. ISO mappings are editorial and not supplied by the census table.

AFG-S011

Institut de Statistiques et d'Études Économiques du Burundi (ISTEEBU/INSBU). (2008). *Répartition de la population issue du RGPH 2008* [official census tables]. https://www.insbu.bi/documents/recensements/R%C3%A9partition-de-la-population-issue-du-RGPH-2008.pdf

Authorship basis: registered authority used as corporate author

Source note: AFG-S011 supplies Population aged 10 years and above by mutually exclusive language read-and-written combination for Burundi. Registered locator: Tableau 3.2, PDF p.62. Registered temporal field: 2008. Caveat: The any-Kirundi count is a derived sum of 16 mutually exclusive total-column rows containing Kirundi. It is nested over the exact Kirundi-only cell; never add the two observations. Counts concern reading and writing among persons aged 10+, not total speakers.

ALG-S007

Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan. (2021). *National composition, religion and language proficiency in the Republic of Kazakhstan* [official national census statistical collection]. https://stat.gov.kz/upload/medialibrary/cee/3rsfg8ps3xo19orb284esg4rx27ihqf7/%D0%9D%D0%B0%D1%86%D0%B8%D0%BE%D0%BD%D0%B0%D0%BB%D1%8C%D0%BD%D1%8B%D0%B9%20%D1%81%D0%BE%D1%81%D1%82%D0%B0%D0%B2.pdf

Authorship basis: registered authority used as corporate author

Source note: ALG-S007 supplies Native language by nationality and population language proficiency for Kazakhstan. Registered locator: Table 6, printed p. 243; Table 8, printed p. 366. Registered temporal field: 2021. Caveat: Table 6 counts language-of-own-nationality only within each nationality and omits cross-nationality native speakers. Table 8 proficiency is not mother tongue or academic comfort; its 17,194,712-person universe appears to be age 5+ from the age table, so the age interpretation is retained as a caveat.

ALG-S008

National Statistical Committee of the Kyrgyz Republic. (2022). *Population and Housing Census of the Kyrgyz Republic 2022: Book II (Part One), Population of Kyrgyzstan* [official national census statistical collection]. https://stat.gov.kg/media/publicationarchive/0f717d2d-5078-4e18-8b56-fb2a530462a2.pdf

Authorship basis: registered authority used as corporate author

Source note: ALG-S008 supplies Native language by ethnic group for Kyrgyzstan. Registered locator: Table 3.8, printed pp. 118-119. Registered temporal field: 2022. Caveat: National Kyrgyz, Uzbek, and Russian native-language totals in the observation file are exact sums of mutually exclusive own-ethnicity and cross-ethnicity cells. Other named groups are nationality-scoped own-language counts and therefore lower bounds on total speakers.

ALG-S009

State Committee of Turkmenistan on Statistics. (2022). *Results of the Complete Population and Housing Census of Turkmenistan - 2022: National Composition and Language Proficiency* [official national census tables]. https://stat.gov.tm/population-census-pdfs/results/en/4.pdf

Authorship basis: registered authority used as corporate author

Source note: ALG-S009 supplies Population by mother tongue for Turkmenistan. Registered locator: Table 4.9, PDF section pp. 8-10. Registered temporal field: 2022. Caveat: Direct mother-tongue totals. Published category labels such as Baloch are retained as-is and do not establish a narrower variety.

ALG-S011

National Institute of Statistics Cambodia. (2019). *General Population Census of the Kingdom of Cambodia 2019: National Report on Final Census Results* [official national census report]. https://nis.gov.kh/nis/Census2019/Final%20General%20Population%20Census%202019-English.pdf

Authorship basis: registered authority used as corporate author

Source note: ALG-S011 supplies Population by mother tongue for Cambodia. Registered locator: Table 2.7.1, PDF p. 53 / printed p. 25. Registered temporal field: 2019. Caveat: The table covers all persons in census households and excludes migrants working abroad. Published total-category counts sum to the stated universe, but multiple male/female subtotals do not sum to their published category totals; only the total-category column is staged.

ALG-S014

General Directorate of Statistics Timor-Leste. (2015). *Timor-Leste Population and Housing Census 2015 Volume 2 Language Tables* [official national census workbook]. https://www.laohamutuk.org/DVD/DGS/Cens15/Census2015v2data.zip

Authorship basis: registered authority used as corporate author

Source note: ALG-S014 supplies Population by mother tongue for Timor-Leste. Registered locator: Workbook 4_2015 V2 Language.xls, sheet 2.12, Table 12. Registered temporal field: 2015. Caveat: The official DGS workbook is preserved from a longstanding external mirror archive. Counts are direct mother-tongue totals; distinct source varieties such as Tetun Prasa and Tetun Terik remain separate.

ALG-S018

National Statistics Committee of the Republic of Uzbekistan. (2026, June 30). *Проведена конференция посвящённая предварительным результатам переписи населения и сельского хозяйства* [official preliminary national census release]. https://stat.uz/ru/press-tsentr/novosti-goskomstata/68978-a-oli-va-ishlo-kh-zhaligini-r-jkhatga-olish-tadbirining-dastlabki-natizhalariga-ba-ishlangan-konferentsiya-tkazildi-3

Authorship basis: registered authority used as corporate author

Source note: ALG-S018 supplies Preliminary national population and Uzbek mother-tongue results for Uzbekistan. Registered locator: Paragraphs reporting the exact preliminary population total and 35.7 million / 91.3 percent Uzbek mother tongue. Registered temporal field: 2026-06-30. Caveat: Preliminary release. The mother-tongue count is rounded to 0.1 million; its separately published 91.3 percent does not arithmetically reproduce 35.7 million against the exact preliminary population total, so the percentage is not used to narrow the count range. Post-enumeration coverage was reported as 97.3 percent.

ASSEC-S006

Transparent CLDR 48.2 target person-count derivations and exclusion audit. (2026, March 17). [transparent secondary derivation witness]. https://raw.githubusercontent.com/unicode-org/cldr/release-48-2/common/supplemental/supplementalData.xml

Authorship basis: title-first reference because the register identifies an arithmetic derivation, not an author

Source note: ASSEC-S006 supplies Nearest-person arithmetic from pinned territory population times pinned language population percentage; no invented confidence interval for Bangladesh; Myanmar; Laos; Mongolia; Sri Lanka; Malaysia; Tajikistan. Registered locator: cldr_target_derivations.md formulas table and exclusion audit. Registered temporal field: 2026-03-17 release; component years not stated. Caveat: Derived point counts inherit CLDR's unspecified component vintages and uncertainty. Integer percentage rounding rules are not published, so observation low/high fields remain blank. Not direct census evidence.

ASSEC-S007

Phạm, B., & McLeod, S. (2016). Consonants, vowels and tones across Vietnamese dialects. *International Journal of Speech-Language Pathology, 18*(2), 122–134. https://doi.org/10.3109/17549507.2015.1101162

Authorship basis: named authors, journal, volume, issue, pages, and DOI read from the hash-verified local PubMed XML witness

Source note: ASSEC-S007 supplies Published lower-bound statement that Vietnamese is spoken by over 89 million people in Vietnam for Vietnam. Registered locator: PMID 27172848; AbstractText label PURPOSE first sentence; volume 18 issue 2 pages 122-134; DOI 10.3109/17549507.2015.1101162. Registered temporal field: 2016; estimate vintage not stated. Caveat: Rounded lower bound rather than a point estimate; L1 versus L2 and source year of the population estimate are not stated. The article is about dialect phonetics rather than population estimation.

ASSEC-S008

CLEAR Global. (2010). *CLEAR Global Language Use Data Platform Philippines national API snapshot* [Weighted census-microdata API]. https://ludp.clearglobal.org/public/location/PHL/?aggregation=0&fields=proportion_value,language_rank,language_name,language_code,location_name,location_code,location_level,dataset_name,url,source,datetime_published,reliability_score,individuals_value_weighted&page_size=500

Authorship basis: CLEAR Global selected from the registered compound authority as the API platform/corporate author; underlying IPUMS Census 2010 provenance remains in the source note

Source note: ASSEC-S008 supplies Weighted persons by normalized main language at national aggregation level zero for Philippines. Registered locator: JSON data rows Cebuano cebu1242; Iloko ilok1237; Hiligaynon hili1240; Waray (Philippines) wara1300; fields individuals_value_weighted and reliability_score. Registered temporal field: 2010-12-31. Caveat: Exact API point values are weighted estimates, not exact headcounts; no variance or confidence interval is exposed. The platform normalizes the metric as main language. API content is mutable, so the local byte witness is controlling.

ASSEC-S018

Transparent Afghanistan Dari and Pashto person-count derivation. (2020). [transparent secondary derivation witness]. https://web.archive.org/web/20230102023706id_/https://www.cia.gov/the-world-factbook/countries/afghanistan/

Authorship basis: title-first reference because the register identifies an arithmetic derivation, not an author

Source note: ASSEC-S018 supplies Nearest-person product of World Bank 2020 population and CIA 2020 total-speaker shares, with official report semantic check for Afghanistan. Registered locator: afghanistan_language_derivation.md formulas for Dari and Pashto. Registered temporal field: 2020. Caveat: Derived point counts inherit the source shares' unknown rounding and uncertainty. No numerical low/high interval is invented. Dari and Pashto totals overlap and must never be summed; displaced cohorts outside Afghanistan are excluded.

GRG-S001

U.S. Census Bureau. (n.d.). *Detailed Languages Spoken at Home and Ability to Speak English for the Population 5 Years and Over in the United States: 2017-2021* [official ACS five-year detailed-language tabulation]. Retrieved August 25, 2026, from https://www2.census.gov/programs-surveys/demo/tables/language-use/2017/2017-2021-acs-lang-tables-nation.xlsx

Authorship basis: registered authority used as corporate author

Source note: GRG-S001 supplies Estimated persons age 5 years and over speaking the named language at home with published 90 percent margin of error for United States. Registered locator: Nation worksheet; selected indented child rows 11, 22, 24, 231, 233-234, 237-239, 246-247, 255, 260, 272, 458, 462, 473-474, 487-488, 492, 508, 510-511, 520, 522, 527, 529, 535, 538, 547-548, 552, 554, 564; notes rows 575-581. Registered temporal field: 2017-2021. Caveat: Sample estimates, not enumerated counts. Universe is age 5 and over and the measure is language spoken at home, not mother tongue or educational access. Parent language-family rows overlap child rows and were excluded. Published margins of error are 90 percent intervals; nonsampling error is not included. Detailed estimates may not sum to aggregates because of rounding.

GRG-S002

Australian Bureau of Statistics. (2021). *2021 Lingiari, Census All persons QuickStats* [official national census QuickStats]. https://www.abs.gov.au/census/find-census-data/quickstats/2021/CED701

Authorship basis: registered authority used as corporate author

Source note: GRG-S002 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Kriol, Djambarrpuyngu, Warlpiri, Murrinh Patha and

Alyawarr rows. Registered temporal field: 2021. Caveat: National values are the Australia comparison column, not the Lingiari local counts. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S003

Australian Bureau of Statistics. (2021). *2021 Alice Springs, Census All persons QuickStats* [official national census QuickStats]. https://www.abs.gov.au/census/find-census-data/quickstats/2021/70201

Authorship basis: registered authority used as corporate author

Source note: GRG-S003 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Pitjantjatjara row. Registered temporal field: 2021. Caveat: National value is the Australia comparison column, not the Alice Springs local count. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S004

Australian Bureau of Statistics. (2021). *2021 Katherine, Census All persons QuickStats* [official national census QuickStats]. https://www.abs.gov.au/census/find-census-data/quickstats/2021/70205

Authorship basis: registered authority used as corporate author

Source note: GRG-S004 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Gurindji, Ngarinyman and Nunggubuyu rows. Registered temporal field: 2021. Caveat: National values are the Australia comparison column, not the Katherine local counts. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S005

Australian Bureau of Statistics. (2021). *2021 Barkly, Census All persons QuickStats* [official national census QuickStats]. https://www.abs.gov.au/census/find-census-data/quickstats/2021/70202

Authorship basis: registered authority used as corporate author

Source note: GRG-S005 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Warumungu row. Registered temporal field: 2021.

Caveat: National value is the Australia comparison column, not the Barkly local count. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S006

Statistics Poland. (2021, March 31). *Size and demographic-social structure in the light of the 2021 Census results* [official national census publication]. https://stat.gov.pl/files/gfx/portalinformacyjny/en/defaultaktualnosci/3701/6/1/1/national_population_and_housing_census_2021_population_2.pdf

Authorship basis: registered authority used as corporate author

Source note: GRG-S006 supplies Persons using the named language at home; respondents could indicate more than one language for Poland. Registered locator: Table 7.5, printed p. 120, PDF p. 122. Registered temporal field: 2021-03-31. Caveat: Multiple language responses were allowed, so rows overlap and must not be summed. The table measures language used at home, not L1 or academic-language comfort. Official bilingual labels are retained exactly. The population source does not report scripts or orthographies.

PM-S001

Office of the Registrar General & Census Commissioner, India. (2011). *C-16 Population by Mother Tongue, Statement 1: Abstract of Speakers' Strength of Languages and Mother Tongues, Census of India 2011* [official census table]. https://censusindia.gov.in/nada/index.php/catalog/42458/download/46089/C-16_25062018.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S001 supplies Persons returned under each census language or mother-tongue category for India. Registered locator: Statement 1, printed pp. 5-12; Part A scheduled-language totals and named mother-tongue components. Registered temporal field: 2011. Caveat: Census language categories can aggregate distinct mother tongues; Hindi parent total overlaps its listed child categories and must not be summed with them. Counts are 2011 observations, not current estimates.

PM-S003

Pakistan Bureau of Statistics. (2023). *Table 11: Population by Mother Tongue, Sex and Rural/Urban, Census 2023* [official census table]. https://www.pbs.gov.pk/wp-content/uploads/census_tables/tables/table_11_national.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S003 supplies All-localities all-sexes persons by reported mother tongue for Pakistan. Registered locator: Table 11, national row, PDF p. 1. Registered temporal field: 2023. Caveat: Several labels are census aggregates (for example Pushto, Balochi, Kohistani); they are not automatically single edition targets.

PM-S004

National Statistics Office, Nepal. (2021). *Languages in Nepal: National Population and Housing Census 2021 Thematic Report* [official census thematic report]. https://censusnepal.cbs.gov.np/results/files/result-folder/Language%20in%20Nepal.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S004 supplies Persons by reported mother tongue for Nepal. Registered locator: Annex 1, printed pp. 123-126. Registered temporal field: 2021. Caveat: Some census labels aggregate varieties; the Sign Language entry is a census mother-tongue response category rather than a full sign-language-use census.

PM-S005

BPS-Statistics Indonesia. (2010). *Kewarganegaraan, Suku Bangsa, Agama, dan Bahasa Sehari-hari Penduduk Indonesia: Hasil Sensus Penduduk 2010* [official census publication]. https://www.bps.go.id/en/publication/2012/05/23/55eca38b7fe0830834605b35/nationality--ethnicity--religion--and-dailylanguage-of-indonesian-population.html

Authorship basis: registered authority used as corporate author

Source note: PM-S005 supplies Population aged 5 and over by language used daily at home for Indonesia. Registered locator: Table L4.1, printed p. 47 (PDF p. 55). Registered temporal field: 2010. Caveat: Home-use counts are not L1 totals. Several rows are broad language groups, including Batak and regional clusters; only individually named cells are retained here.

Local evidence status: No local witness path or hash is registered; do not describe this source as locally hash-verified.

PM-S007

Instituto Nacional de Estadística e Informática (INEI). (2017). *Peru: Perfil Sociodemográfico, Censos Nacionales 2017* [official census report]. https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1544/00TOMO_01.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S007 supplies Census population aged 3 and over by language or mother tongue learned in childhood for Peru. Registered locator: Cuadro 4, printed p. 618, total column. Registered temporal field: 2017. Caveat: Quechua and Aimara are aggregate census labels spanning varieties. The total column is used from a health-insurance cross-tabulation; it remains the national count for the stated age and language universe.

PM-S011

Zimbabwe National Statistics Agency (ZIMSTAT). (2022). *Zimbabwe 2022 Population and Housing Census Report, Volume 1* [official census report]. https://www.zimstat.co.zw/wp-content/uploads/Demography/Census/2022_PHC_Report_27012023_Final.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S011 supplies Population by reported mother tongue for Zimbabwe. Registered locator: Table 2.17, printed p. 123. Registered temporal field: 2022. Caveat: Census labels do not resolve every dialect or orthographic choice. Sign Language is a census mother-tongue category, not a complete sign-language-use measure.

PM-S013

Statistics Mauritius. (2022). *2022 Housing and Population Census, Volume II: Demographic and Fertility Characteristics* [official census report]. https://statsmauritius.govmu.org/Documents/Census_and_Surveys/Census2022/HPC_TR_Vol2_Demography_Yr22.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S013 supplies Resident population by language usually spoken at home for Mauritius. Registered locator: Table D9, printed p. 165, Republic of Mauritius row. Registered temporal field: 2022. Caveat: Chinese languages is an aggregate. Home language is not necessarily L1. The published table excludes Agalega and St Brandon as documented in the report.

PM-S014

Stats NZ. (2023). *Place and ethnic group summaries: Total New Zealand population language comparator* [official census summary table]. https://tools.summaries.stats.govt.nz/ethnic-group/pacific-peoples-not-elsewhere-classified

Authorship basis: registered authority used as corporate author

Source note: PM-S014 supplies Persons who reported speaking each language; multiple-response census measure for New Zealand. Registered locator: Table 'Percentage of population that speak each language', Total New Zealand population, 2023 column. Registered temporal field: 2023. Caveat: Multiple responses are permitted and figures use fixed random rounding; rows are not L1 counts and must not be summed. Northern Chinese is a census category, not a single unqualified language target.

Local evidence status: No local witness path or hash is registered; do not describe this source as locally hash-verified.

PM-S016

Central Statistics Office Ireland. (2022). *Census of Population 2022 Summary Results: Education and Irish Language* [official census release]. https://www.cso.ie/en/releasesandpublications/ep/p-cpsr/censusofpopulation2022-summaryresults/educationandirishlanguage/

Authorship basis: registered authority used as corporate author

Source note: PM-S016 supplies Population aged 3 and over reporting ability to speak Irish; and daily use outside education for Ireland. Registered locator: Table 6.2 and Figure 6.4 text. Registered temporal field: 2022. Caveat: Ability and daily-use measures are not interchangeable; cells are separated and must not be summed.

Local evidence status: No local witness path or hash is registered; do not describe this source as locally hash-verified.

PM-S020

Central Statistical Agency of Ethiopia (now Ethiopian Statistical Service). (2007). *Population and Housing Census 2007: National Statistical Report* [official census statistical report]. https://ess.gov.et/wp-content/uploads/2007/09/Population-and-Housing-Census-2007-National_Statistical.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S020 supplies Population by mother tongue used with family members or guardians during childhood for Ethiopia. Registered locator: Table 3.2, printed pp. 91-93, Urban + Rural both sexes. Registered temporal field: 2007. Caveat: The observation is old but remains the latest Ethiopian national census. Several census labels, especially Guragiegna, aggregate distinct languages; counts are not current projections.

TH-S001

National Statistical Office Thailand. (2010, September 1). *The 2010 Population and Housing Census (Whole Kingdom)* [official national census report]. https://catalogapi.nso.go.th/api/doc/department/D10/SD10_04/SD10_04_165_2.pdf

Authorship basis: registered authority used as corporate author

Source note: TH-S001 supplies Population by usual languages spoken at home for Thailand. Registered locator: Table 7, PDF pp. 98-99; census day described on PDF pp. 49-50. Registered temporal field: 2010-09-01. Caveat: The table mixes mutually exclusive Thai-only/Thai-plus-other/other-only strata with named other-language categories that can overlap the Thai-plus-other stratum. English labels contain several typographic or country-name substitutions. Counts are 2010 observations, not current projections; the report warns that independently rounded totals may differ slightly.

ZA-S001

Statistics South Africa. (n.d.). *Census 2022 in Brief* [official national census report]. Retrieved August 25, 2026, from https://www.statssa.gov.za/publications/Census2022inBrief/Census2022inBriefJune2024.pdf

Authorship basis: registered authority used as corporate author

Source note: ZA-S001 supplies Population aged 1 year and older by language most often spoken in the household for South Africa. Registered locator: Table 3.8 (1)-(2), printed pp. 26-27; PDF pp. 34-35. Registered temporal field: 2024-06. Caveat: The measure is language most often spoken with other household members, not mother tongue, proficiency, literacy, or academic-language comfort. The table excludes Not applicable (638,736) and Unspecified (258,967). Census label 'Sign language' does not identify a signed-language variety.

Profile and orthography authorities

S-ACS

U.S. Census Bureau. (n.d.). *Detailed Languages Spoken at Home and Ability to Speak English for the Population 5 Years and Over in the United States: 2017-2021* [official census workbook]. Retrieved August 25, 2026, from https://www2.census.gov/programs-surveys/demo/tables/language-use/2017/2017-2021-acs-lang-tables-nation.xlsx

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-ACS is profile/orthography authority for the interventions listed here. Exact registered evidence: Nation worksheet supplies the retained point estimates and 90% margins of error; universe is age 5+ language spoken at home. Authority level: primary national statistical authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-APW

Dilzhe'e Apache Dictionary. (n.d.). *The Pronunciation and Spelling of Dilzhe'e Apache* [community dictionary orthography]. Retrieved August 25, 2026, from https://dilzhee.western-apache.org/Spelling

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-APW is profile/orthography authority for the interventions listed here. Exact registered evidence: Defines the dictionary spelling and its explicit changes from White Mountain and San Carlos Western Apache practice. Authority level: community language-resource authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CH-GU

Kumision i Fino' CHamoru yan i Fina'naguen i Historia yan i Lina'la' i Taotao Tano'. (n.d.). *Utugrafihan CHamoru Guahan* [official orthography]. Retrieved August 25, 2026, from https://kumisionchamoru.guam.gov/wp-content/uploads/2024/07/utugrafihan_chamoru_guahan.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CH-GU is profile/orthography authority for the interventions listed here. Exact registered evidence: Complete adopted 2024 Guam CHamoru orthography. Authority level: Guam statutory language commission. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CH-MP

Inetnun Kutturan Natibun Marianas / Kkoor Aramasal Marianas. (n.d.). *Revised Chamorro-English Dictionary introduction* [community dictionary documentation]. Retrieved August 25, 2026, from https://natibunmarianas.org/dictionary-introduction/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CH-MP is profile/orthography authority for the interventions listed here. Exact registered evidence: States there are two standard orthographies and that all headwords use the official CNMI orthography adopted in 2010. Authority level: CNMI community language institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CHO-MS

Mississippi Band of Choctaw Indians. (n.d.). *Mississippi Band of Choctaw Indians 2023 Tribal Profile* [official tribal publication]. Retrieved August 25, 2026, from https://www.choctaw.org/wp-content/uploads/2024/01/2023-Tribal-Profile_v19-updated.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CHO-MS is profile/orthography authority for the interventions listed here. Exact registered evidence: Describes the modern Mississippi Choctaw alphabet as a variant of the Byington alphabet and documents tribal language programming. Authority level: tribal government. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CHO-OK

Choctaw Nation of Oklahoma School of Choctaw Language. (n.d.). *Choctaw alphabet* [official tribal-language reference]. Retrieved August 25, 2026, from https://dictionary.choctawnation.com/alphabet/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CHO-OK is profile/orthography authority for the interventions listed here. Exact registered evidence: Defines the Choctaw Nation instructional alphabet. Authority level: tribal government language authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-DAKOTA-DIO

Dakhota Iapi Okhodakichiye. (n.d.). *About Dakhota Iapi Okhodakichiye* [community language institution]. Retrieved August 25, 2026, from https://dakhota.org/about-dio/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-DAKOTA-DIO is profile/orthography authority for the interventions listed here. Exact registered evidence: Delimits the organization to Eastern/Santee-Sisseton Dakota language work. Authority level: community authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-DAKOTA-SWC

Sisseton Wahpeton College. (n.d.). *Dakota Dictionary introduction* [tribal-college dictionary documentation]. Retrieved August 25, 2026, from https://dakotadictionary.hartman-technology.com/uploads/content/Introduction.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-DAKOTA-SWC is profile/orthography authority for the interventions listed here. Exact registered evidence: Documents the University of Minnesota orthography used at Sisseton Wahpeton College and contrasts another system. Authority level: tribal college. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-GARIFUNA-ALPH

Honduras Secretaria de las Culturas las Artes y los Patrimonios. (n.d.). *Alfabeto Garifuna* [official alphabet]. Retrieved August 25, 2026, from https://arc.secapph.gob.hn/wp-content/uploads/2025/10/Alfabeto-Garifuna.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-GARIFUNA-ALPH is profile/orthography authority for the interventions listed here. Exact registered evidence: Publishes the alphabet used for the Honduras educational profile. Authority level: national government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-GARIFUNA-HN

Honduras Secretaria de las Culturas las Artes y los Patrimonios. (n.d.). *Unidad de Educacion Plurilingue y Multicultural* [official government education page]. Retrieved August 25, 2026, from https://arc.secapph.gob.hn/direccion-general-de-las-culturas-y-los-patrimonios/unidad-de-educacion-plurilingue-y-multicultural

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-GARIFUNA-HN is profile/orthography authority for the interventions listed here. Exact registered evidence: Documents government production of Garifuna educational texts and links the official alphabet. Authority level: national government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-HOPI

Hopi Dictionary Project. (n.d.). *Learn Hopi / Third Mesa dictionary materials* [institutional teaching and dictionary resource]. Retrieved August 25, 2026, from https://hopidictionary.com/wp-content/uploads/2024/10/learnhopi.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-HOPI is profile/orthography authority for the interventions listed here. Exact registered evidence: Provides a reproducible Third Mesa-oriented Hopi writing and teaching profile. Authority level: community and University of Arizona project. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-IANA

Internet Assigned Numbers Authority. (n.d.). *Language Subtag Registry* [standards registry]. Retrieved August 25, 2026, from https://www.iana.org/assignments/language-subtag-registry/language-subtag-registry

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-IANA is profile/orthography authority for the interventions listed here. Exact registered evidence: Validates emitted language/script/region components and identifies ik as a macrolanguage with esi and esk members. Authority level: primary international registry. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-INUPIAQ

Alaska Native Language Archive / University of Alaska Fairbanks. (n.d.). *Guide to the Inupiaq Language Collection* [public-university archive guide]. Retrieved August 25, 2026, from https://www.uaf.edu/anla/collections/inupiaq/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-INUPIAQ is profile/orthography authority for the interventions listed here. Exact registered evidence: Identifies two major groups and four dialect types and records an accepted 1970s orthography. Authority level: public university. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-INUPIAQ-NS

University of Alaska Fairbanks Documenting Alaskan and Neighboring Languages. (n.d.). *North Slope Inupiaq lexicon* [public-university lexicon record]. Retrieved August 25, 2026, from https://www.uaf.edu/danl/project-updates/edna-maclean/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-INUPIAQ-NS is profile/orthography authority for the interventions listed here. Exact registered evidence: Identifies the completed lexicon as the definitive dictionary for North Slope Inupiaq. Authority level: public university. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-LKT

Lakota Language Consortium. (n.d.). *Lakota Stories and Lakota Level 1 textbook* [institutional teaching and reader resources]. Retrieved August 25, 2026, from https://lakhota.org/product/lakota-stories-collected-by-ella-deloria/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-LKT is profile/orthography authority for the interventions listed here. Exact registered evidence: Defines a consistent contemporary phonemic orthography used in the reader and a pedagogically consistent textbook series. Authority level: language-teaching institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-ODAWA-DICT

Algonquian Dictionaries Project. (n.d.). *Nishnaabemwin Odawa and Eastern Ojibwe online dictionary* [institutional dictionary]. Retrieved August 25, 2026, from https://dictionary.nishnaabemwin.atlas-ling.ca/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-ODAWA-DICT is profile/orthography authority for the interventions listed here. Exact registered evidence: Identifies an exact Odawa/Eastern Ojibwe dictionary data edition suitable as a named production profile. Authority level: community and academic dictionary project. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-ODAWA-TRIBAL

Adawe Cultural Center / Ottawa Tribe of Oklahoma. (n.d.). *Language resources* [official tribal cultural resource]. Retrieved August 25, 2026, from https://www.heritage.ottawatribe.gov/language

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-ODAWA-TRIBAL is profile/orthography authority for the interventions listed here. Exact registered evidence: Corroborates tribal use and recommendation of Double-Vowel language resources. Authority level: tribal cultural institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-OJ

University of Minnesota Ojibwe People's Dictionary. (n.d.). *About the Ojibwe language* [institutional dictionary documentation]. Retrieved August 25, 2026, from https://ojibwe.lib.umn.edu/about-ojibwe-language

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-OJ is profile/orthography authority for the interventions listed here. Exact registered evidence: Names the Central Southwestern variety and states that the dictionary uses the Double-Vowel system. Authority level: public-university and community consortium. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PAP-AW

Government of Aruba. (n.d.). *First Bible in Papiamento in Aruba's official spelling* [official government language page]. Retrieved August 25, 2026, from https://www.gobierno.aw/en/first-bible-in-papiamento-in-arubas-official-spelling

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PAP-AW is profile/orthography authority for the interventions listed here. Exact registered evidence: Explicitly distinguishes Aruba's official spelling from Curaçao's existing Papiamentu edition. Authority level: territorial government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PAP-CW

Government of Curaçao. (n.d.). *Landsbesluit schrijfwijze Papiamentu en Nederlands* [official spelling decree]. Retrieved August 25, 2026, from https://gobiernu.cw/wp-content/uploads/2025/12/196-GT.-Lb-schrijfwijze-Papiamentu-en-Nederlands.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PAP-CW is profile/orthography authority for the interventions listed here. Exact registered evidence: Establishes the official Papiamentu spelling and wordlist used in Curaçao; the same phonological tradition is the statutory Bonaire baseline. Authority level: territorial government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PAP-FACT

United Kingdom Foreign Commonwealth and Development Office. (n.d.). *Kingdom of the Netherlands Toponymic Factfile* [official government factfile]. Retrieved August 25, 2026, from https://assets.publishing.service.gov.uk/media/68419e2efa0289a17a3e879c/Kingdom_of_the_Netherlands_Toponymic_Factfile.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PAP-FACT is profile/orthography authority for the interventions listed here. Exact registered evidence: Distinguishes Aruba Papiamento from Curaçao/Bonaire Papiamentu usage. Authority level: government institutional source. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PDC

Deitsh Books. (n.d.). *Pennsylvania German spelling-system resources* [community publishing resource]. Retrieved August 25, 2026, from https://www.deitshbooks.com/tools/books/grammer/

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PDC is profile/orthography authority for the interventions listed here. Exact registered evidence: States that two written spelling systems exist and identifies the English-oriented system used in named books. Authority level: community language publisher. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-YIVO

YIVO Institute for Jewish Research. (n.d.). *Basic Facts about Yiddish* [institutional language reference]. Retrieved August 25, 2026, from https://www.yivo.org/cimages/basic_facts_about_yiddish_2014.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-YIVO is profile/orthography authority for the interventions listed here. Exact registered evidence: Section on Standard Yiddish defines the YIVO literary standard while delimiting it from any single spoken dialect. Authority level: major cultural and scholarly institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.